

BML: Brief Background & Graphical Modelling

Derek Aguiar

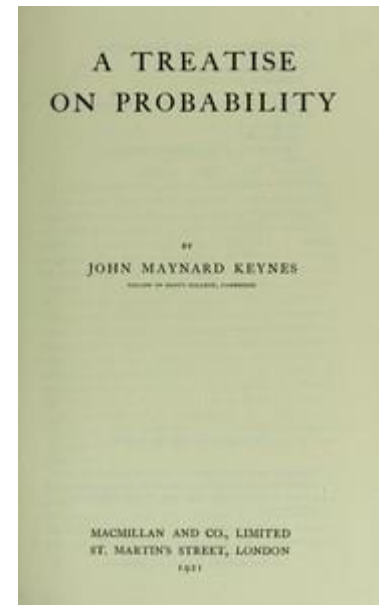
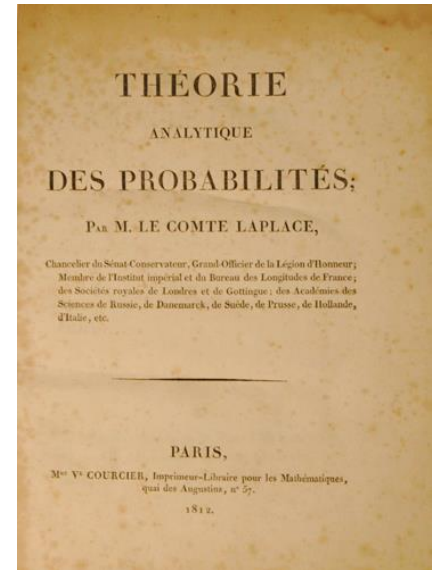
University of Connecticut – CSE5825

Announcements

- Background + directed and undirected graphical models today
- Readings have been up
- Next week readings will be up tonight
- Start thinking about problems in your area of interest

Probability

- Mathematical formalization of uncertainty
- “Life’s most important problems are, for the most part, problems in probability.”
 - Laplace *Analytic Theory of Probability* 1812
- “the probable is the hypothesis on which it is rational for us to act”
 - Keynes *A treatise on probability* 1921



Probability

- In BML, we are often interested in the probability of certain events

$$p(\text{Friday} = \text{Snow})$$

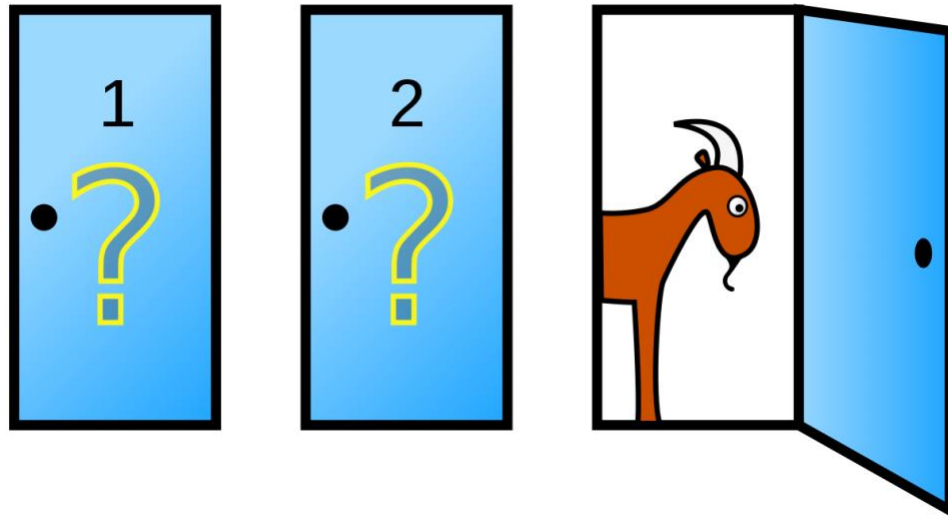


$$p(\text{call} = \text{spam})$$



A motivating problem

- Three doors, prize behind random door, goats behind others
- You select a door, the host reveals a goat behind a different door
- Should you stick with your original pick? Or switch?



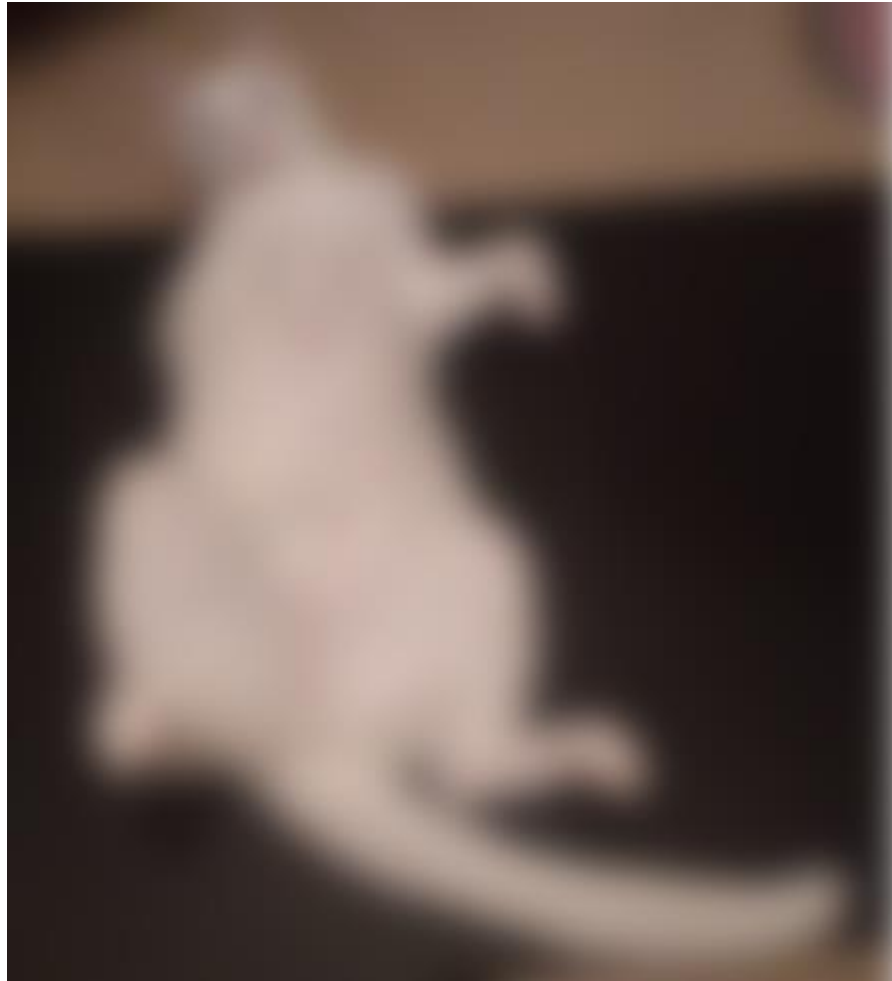
Inference Problem

- We are presented with a problem with unknowns
- We wish to model these unknowns with probabilities and reason about their likely values
- Inference
 - ‘to derive as a conclusion from facts or premises’ - MW
 - make conclusions about values of latent (not observed) random variables given observations of related random variables

Inference Problem: Fight or flight?

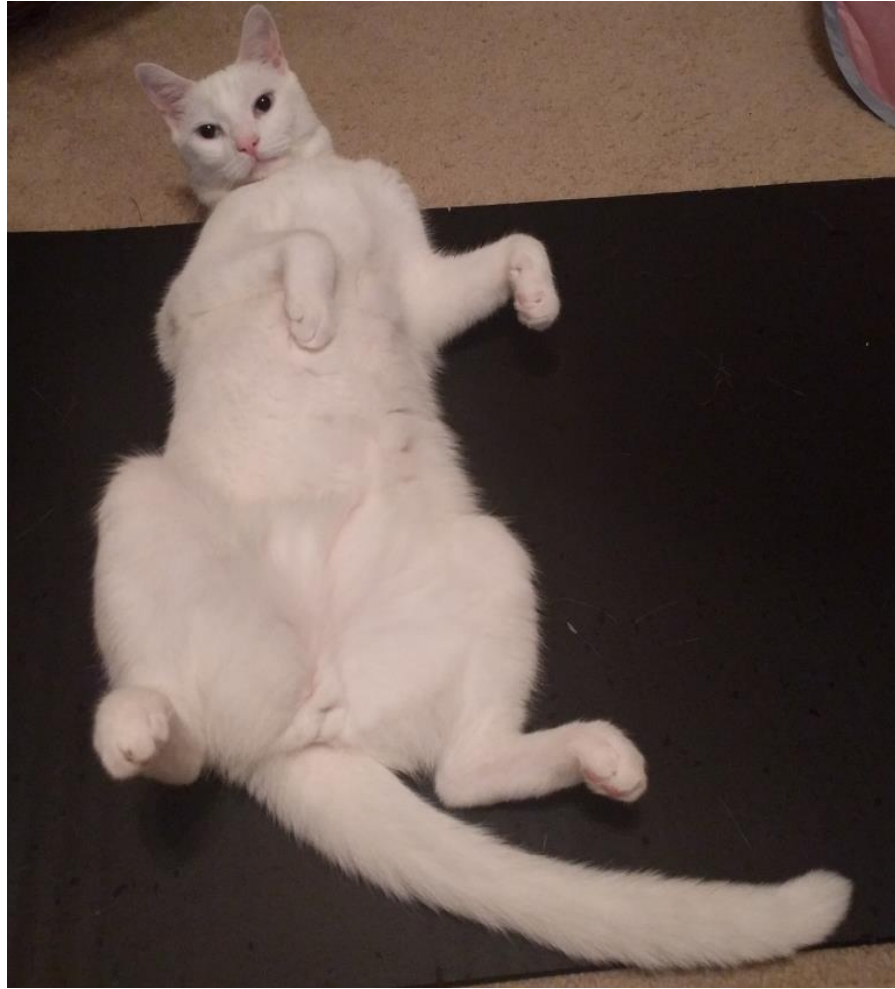


Inference Problem: Fight or flight?



Investigate?
Or run?

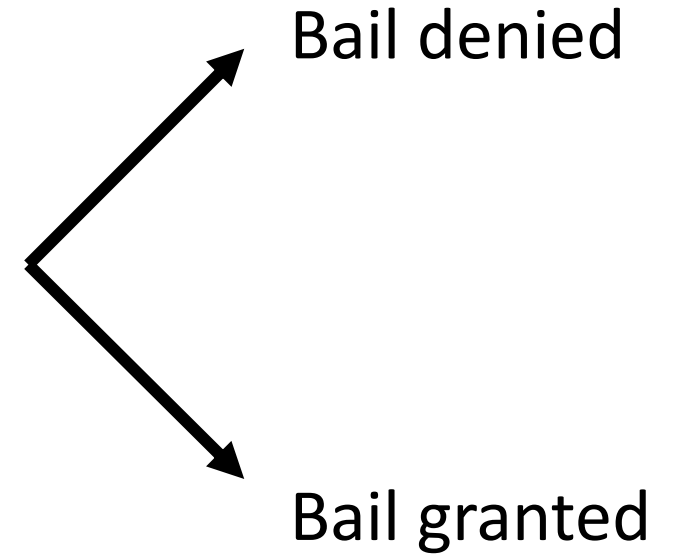
Inference Problem: Fight or flight?



Inference Problem in Law: Denying Bail



Facts of the case
Accused
background
External factors



Inference Problem in Medicine: Drug Assignment



Patient Features
Drug Features
Diagnostics

Which drug to
administer?



These are all questions about probabilities

Random variables

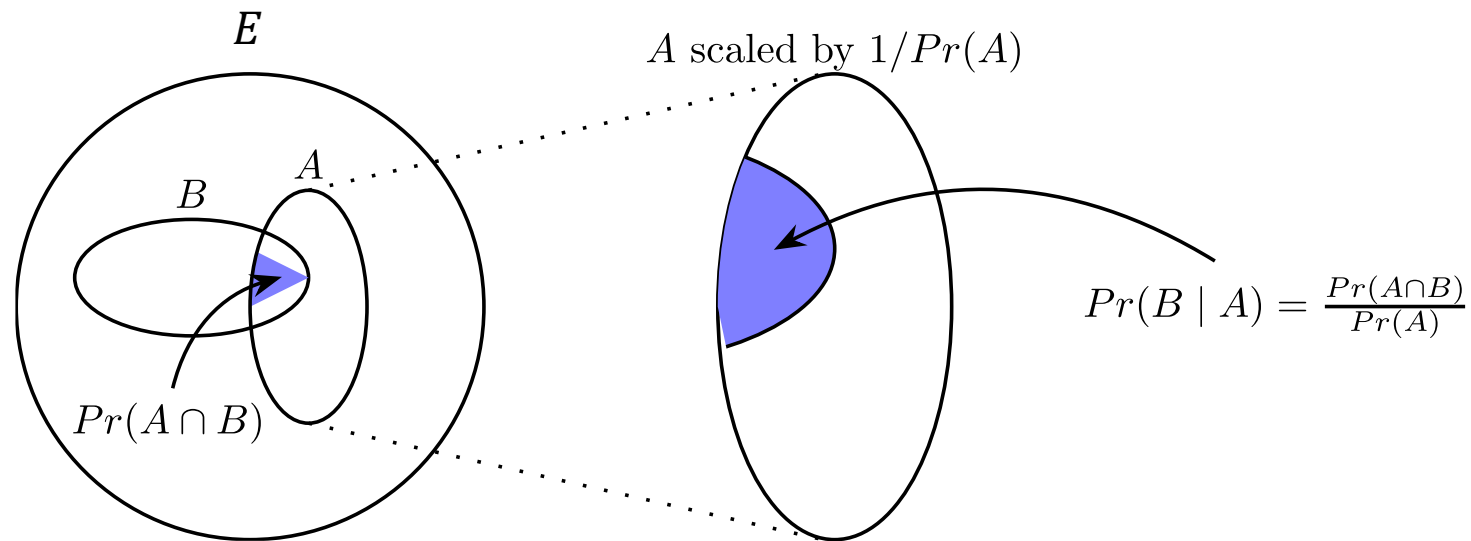
- In all proceeding problems, we need some mathematical construct to describe random events: *random variables*
 - E.g. Will a flipped coin come up heads or tails?
- Express values in a *sample space* Ω : set of outcomes.
- Values of discrete random variables are *atoms*
- *Discrete probability distributions* assign probability to each atom in the sample space

Random variables

- *events* $\mathcal{F}$: a set whose elements $A \in \mathcal{F}$ are subsets of Ω (atoms)
- A **random variable** is a function mapping the sample space to a probability.
- A **probability measure** is a function $P: \mathcal{F} \rightarrow \mathbb{R}$ that satisfies:

Joint and conditional

- If we have a collection of random variables, the *joint distribution* is a distribution over their configurations
- Conditional distribution is the distribution of a subset of random variables when another subset is known.



Chain Rule

Marginalization

- We will often define variables that make sense to include in the model (they capture some form of uncertainty) but we do not care about their particular values

Bayes Rule

Bayes Rule

posterior

data likelihood

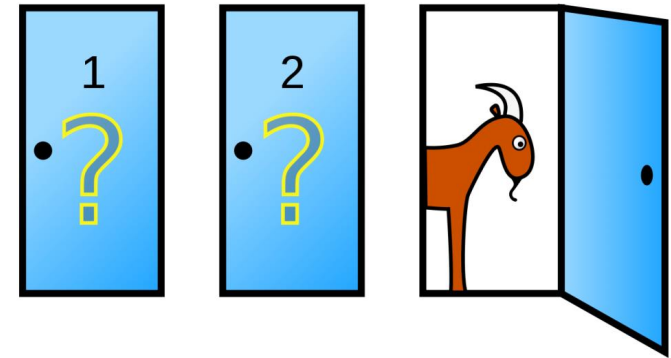
prior

$$P(Y|X) = \frac{P(X|Y)P(Y)}{\sum_y P(X|Y = y)P(Y = y)}$$

marginal likelihood
or model evidence

Monty Hall

- W.l.o.g, assume you select door 1
- A_x = event that host opens door x
- B_x = prize behind door x



Independence

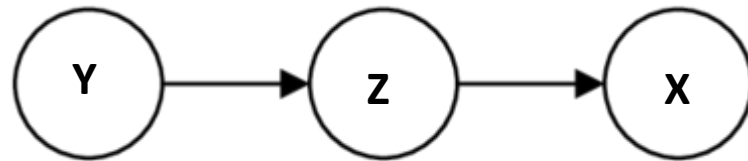
$$P(Y|X) = P(Y)$$

$$X \perp\!\!\!\perp Y \Leftrightarrow P(X, Y) = P(Y|X)P(X) = P(X|Y)P(Y) = P(X)P(Y)$$

Conditional Independence

- X and Y are conditionally independent given Z if

$$P(X|Y, Z = z) = P(X|Z = z)$$

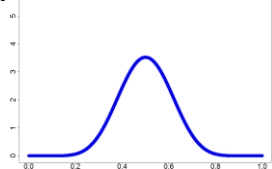


Probabilistic ML workflow

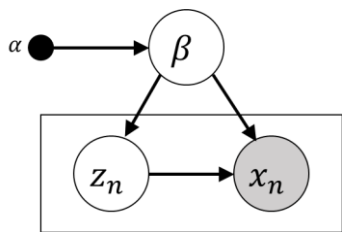
domain expertise



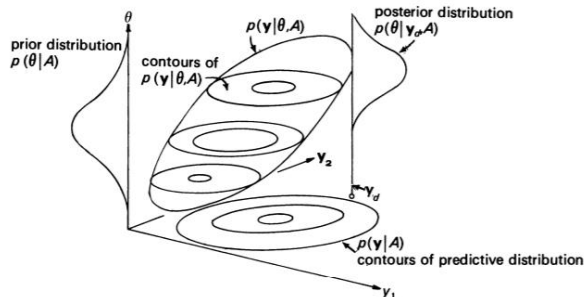
prior knowledge



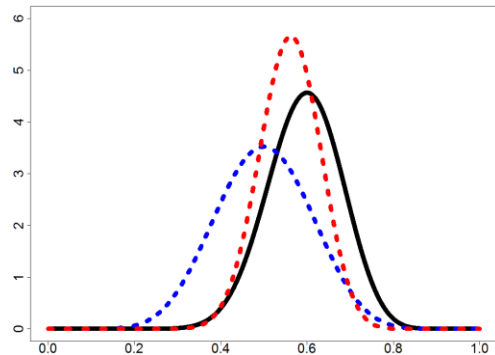
Probabilistic modeling



model criticism



Posterior inference

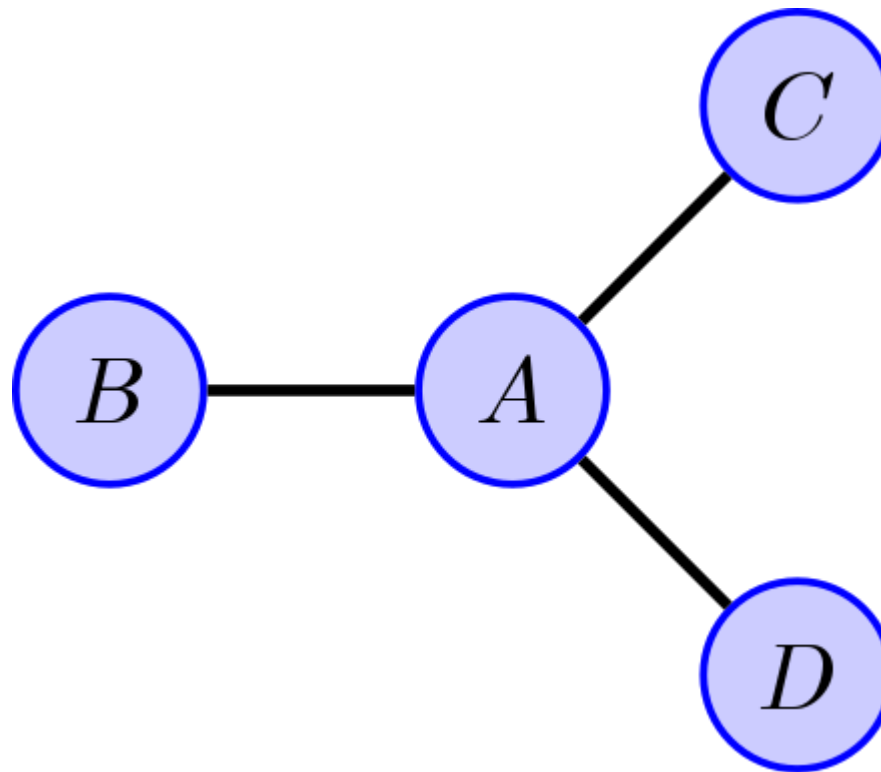
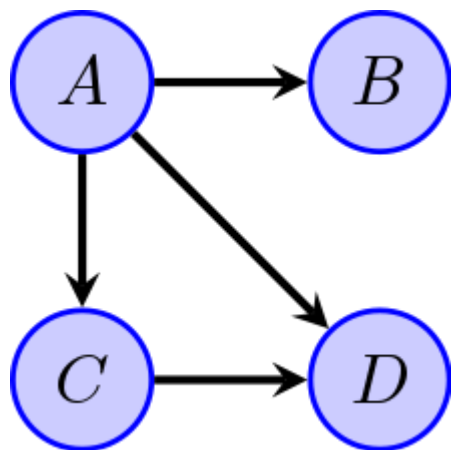


predictions & exploration

CGACCCTCCAGTGGAGCCAG
CGACCCTCCAGTGGAGCCAG
CGACACTATCAATAGAGCCAG
CGACACTATCAATAGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGAACGAGGCAGA
AACTCTGATGAACGGAAACAG

data

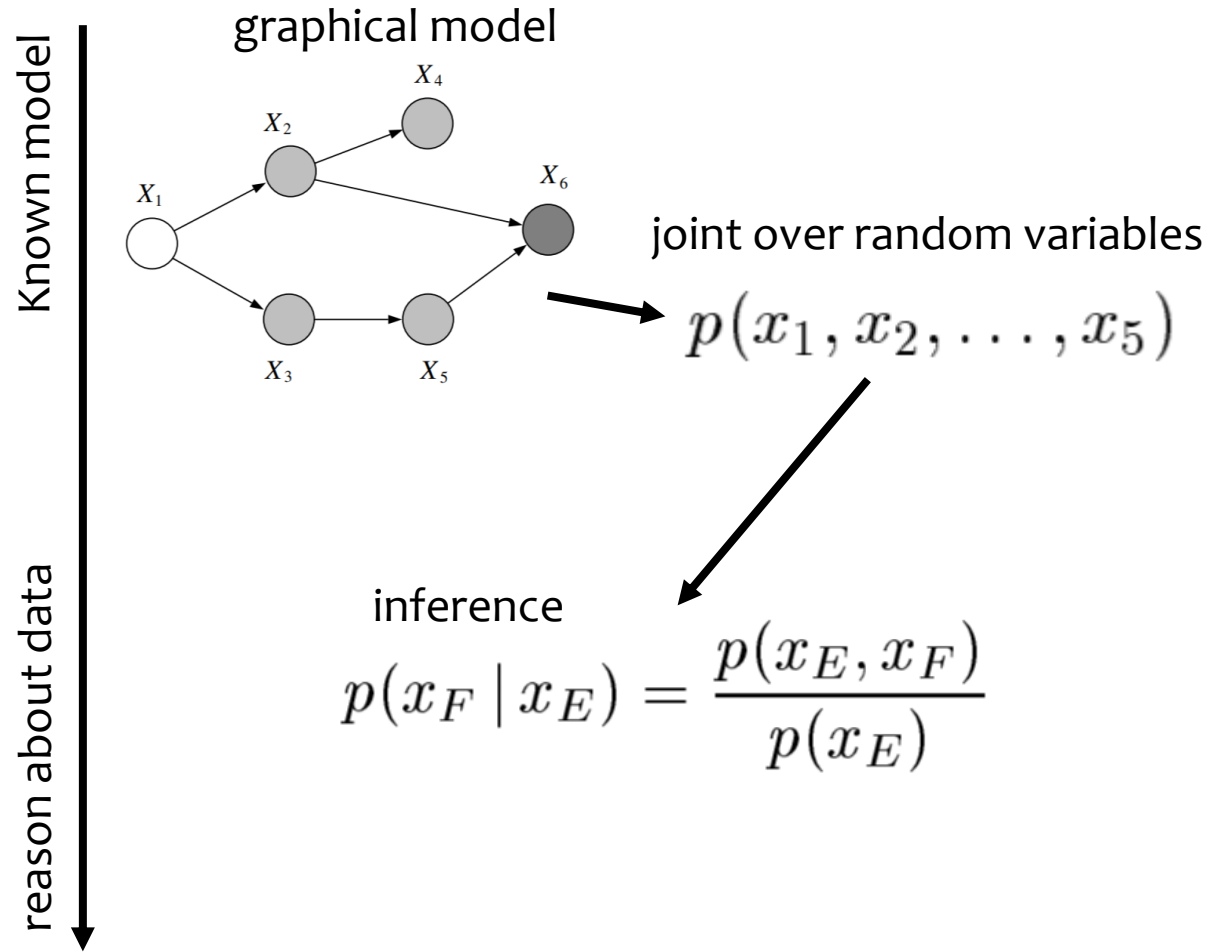
CGACCCTCCAGTGGAGCCAG
CGACCCTCCAGTGGAGCCAG
CGACACTATCAATAGAGCCAG
CGACACTATCAATAGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGAACGAGGCAGA
AACTCTGATGAACGGAAACAG



Graphical Models

Directed and undirected

Probabilist



Motivation – Correlations among variables

- Suppose we observe several correlated variables $\{X_1, \dots, X_n\}$



CTCTTGGCTCCAGCATCGATGAAGAACGCA
ATTTAGAGGAAGTAAAAGTCGTAACAAGGT
ACTGTCAAAACTTTAAACAACGGATCTCTT
TTGCTTCGGCGGGCGCCCGCAAGGGTGCCCC
CCTGCCGTGGCAGATCCCCAACGCCGGGCC
TCTTGGCTCCAGCATCGATGAAGAACGCA
GCATCGATGAAGAACGCAGCGAAACGCGAT
ATACTTCTGAGTGTTCTTAGCGAACTGTCA
GATCTCTTGGCTCCAGCATCGATGAAGAAC

Motivation – Correlations among variables

- Which variables are independent?
 - Are DNA at positions 1 and 1000 distributed independently?
- Which variables are conditionally independent given others?
 - Can I ignore the pixels around Homer's mouth when calculating the distribution of pixel colors in his eyes?
- What are the marginal distributions of subsets of variables?
 - What is the probability of a “G” at this probable disease location on the genome?

GATAGATATAGCGATA

2011 Turing Award on GM



MORE ACM AWARDS



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ALPHABETICAL LISTING

YEAR OF THE AWARD

RESEARCH SUBJECT



[Photo-Essay](#)

BIRTH:
September 4, 1936, Tel Aviv.

EDUCATION:
B.S., Electrical Engineering (Technion, 1960); M.S., Electronics (Newark College of Engineering, 1961); M.S., Physics (Rutgers University, 1965); Ph.D., Electrical Engineering (Polytechnic Institute of Brooklyn, 1965).

EXPERIENCE:
Research Engineer, New York University Medical School (1960–1961); Instructor,

JUDEA PEARL
United States – 2011

CITATION
For fundamental contributions to artificial intelligence through the development of a calculus for probabilistic and causal reasoning.


SHORT ANNOTATED BIBLIOGRAPHY


ACM DL AUTHOR PROFILE


ACM TURING AWARD LECTURE VIDEO


RESEARCH SUBJECTS


ADDITIONAL MATERIALS

Judea Pearl created the representational and computational foundation for the processing of information under uncertainty.

He is credited with the invention of [Bayesian networks](#), a mathematical formalism for defining complex probability models, as well as the principal algorithms used for inference in these models. This work not only revolutionized the field of artificial intelligence but also became an important tool for many other branches of engineering and the natural sciences. He later created a mathematical framework for [causal inference](#) that has had significant impact in the social sciences.

Judea Pearl was born on September 4, 1936, in Tel Aviv, which was at that time administered under the British Mandate for Palestine. He grew up in [Bnei Brak](#), a Biblical town his grandfather went to reestablish in 1924. In 1956, after serving in the Israeli army and joining a Kibbutz, Judea decided to study engineering. He attended the Technion, where he met his wife, Ruth, and received a B.S. degree in Electrical Engineering in 1960. Recalling the Technion faculty members in a 2012 interview in the *Technion Magazine*, he emphasized the thrill of discovery:

Graphical Model Interpretation

- Graphical models can be thought of as probabilistic database, a machine that can answer queries regarding the values of random variables
- We build the database in pieces using probability theory to ensure the pieces have a consistent interpretation
- This table represents the joint probability distribution; it captures many of the queries that we want

	H	T
H	0.25	0.25
T	0.25	0.25

Naïve representation

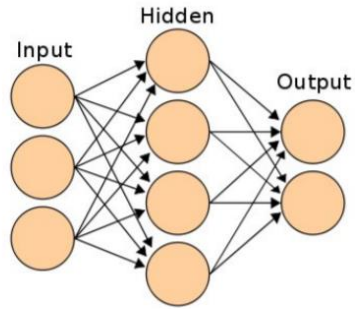
- Assuming n variables, each with r values
 - How big is the lookup table?

Naïve representation

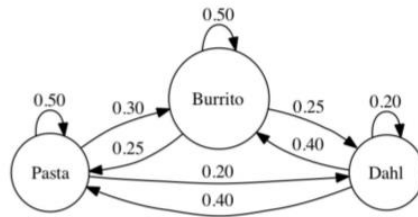
- Assuming n variables, each with r values
 - r^n lookup table (exponential storage)
 - Hard to interpret (e.g., conditional independence)
- Graphical modelling is the study of probabilistic models
 - All probabilistic models can be represented as graphs, but not all graphs are probabilistic models
 - Graphical models must specify:
 - **Random variables**
 - **Conditional independencies**

Naïve representation

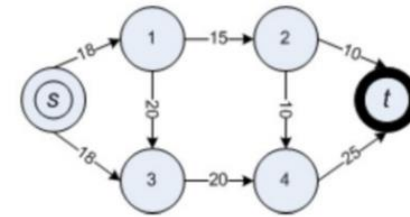
- Which of the following are probabilistic graphical models?



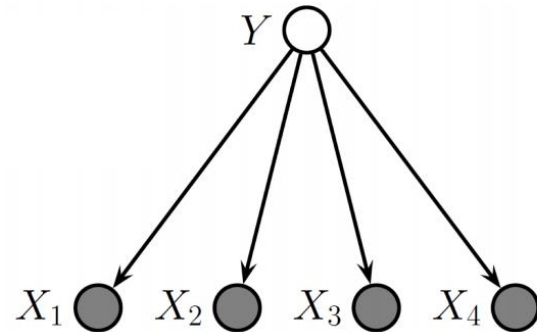
neural network



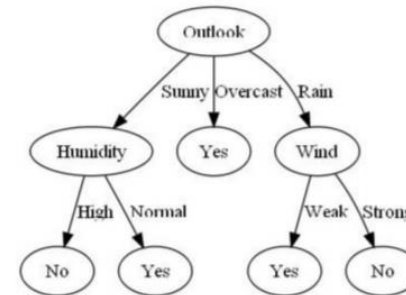
HMM template



network flow



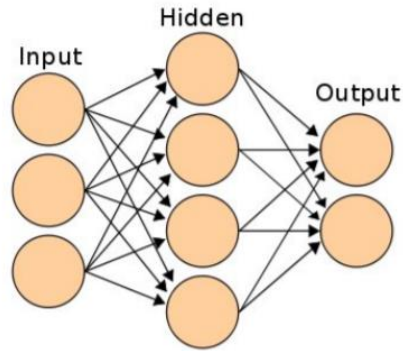
naïve Bayes



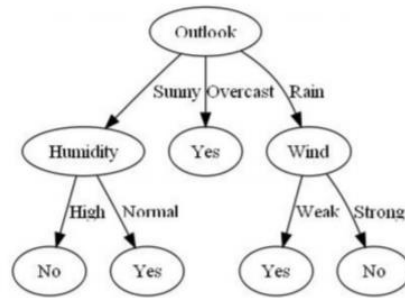
decision tree

Naïve representation

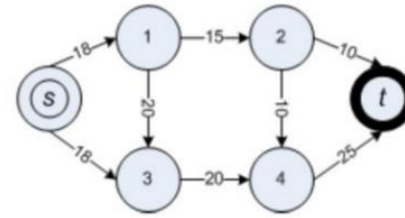
These are not graphical models



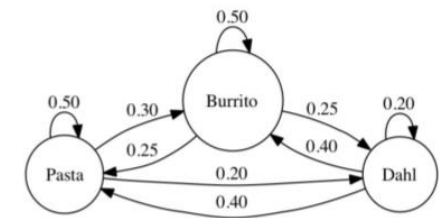
neural network



decision tree



network flow



HMM template

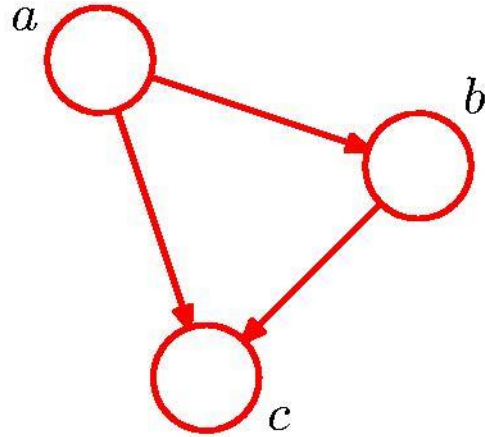
- There are no explicit random variables
- Some probabilistic interpretations of each!

Notation

- Random variables $\{X_1, \dots, X_n\}$
- x_i realization of X_i
- queries about independence of r.v.s, conditional probabilities can be answered if we have the joint $P(X_1 = x_1, \dots, X_n = x_n)$
- To simplify, we will represent probabilities in terms of PMF
 $p(x_1, x_2, \dots, x_n) \triangleq P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$
- Also, $x = \{x_1, x_2, \dots, x_n\}$ and $X = \{X_1 = x_1, X_2 = x_2, \dots, X_n = x_n\}$

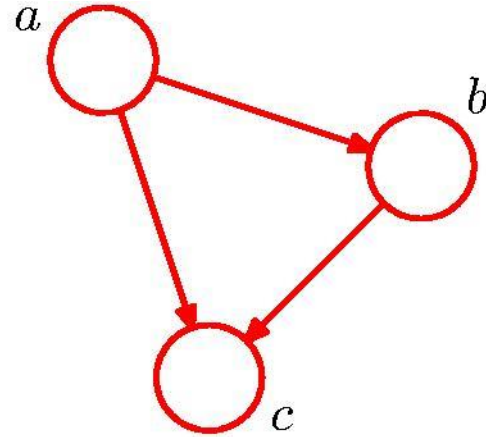
Directed graphical models

- Let $G(V, E)$ be an acyclic graph where V is a set of nodes and E is a set of directed edges



Directed graphical models

- Let π_i denote the parents of V_i and X_{π_i} the r.v.s associated with parents of V_i .



- We associate each node with a function $f_i(x_i, x_{\pi_i})$ assumed to have properties of conditional distributions

Directed graphical models

- Let $V = \{1, 2, \dots, n\}$ and given functions $\{f_i(x_i, x_{\pi_i}) : i \in V\}$, the joint probability is defined as

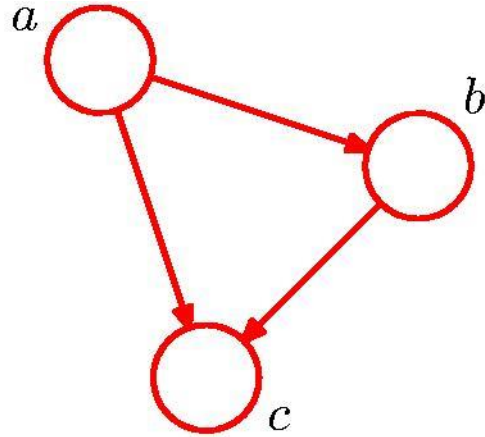
$$p(x_1, x_2, \dots, x_n) \triangleq \prod_{i=1}^n f_i(x_i, x_{\pi_i}).$$

- Can show the functions $f_i(x_i, x_{\pi_i})$ are exactly $p(x_i | x_{\pi_i})$, so

$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p(x_i | x_{\pi_i}).$$

Factorization using Directed Graphical Models

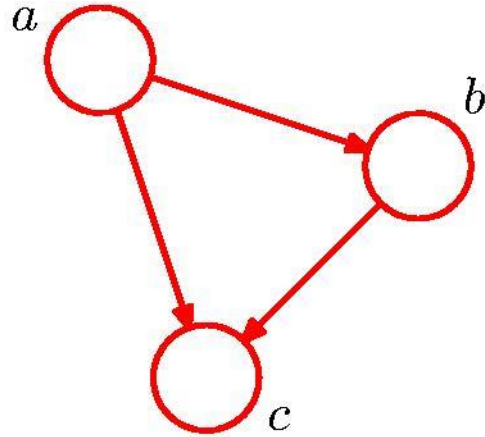
What is $p(a, b, c)$?



$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p(x_i \mid x_{\pi_i}).$$

Factorization using Directed Graphical Models

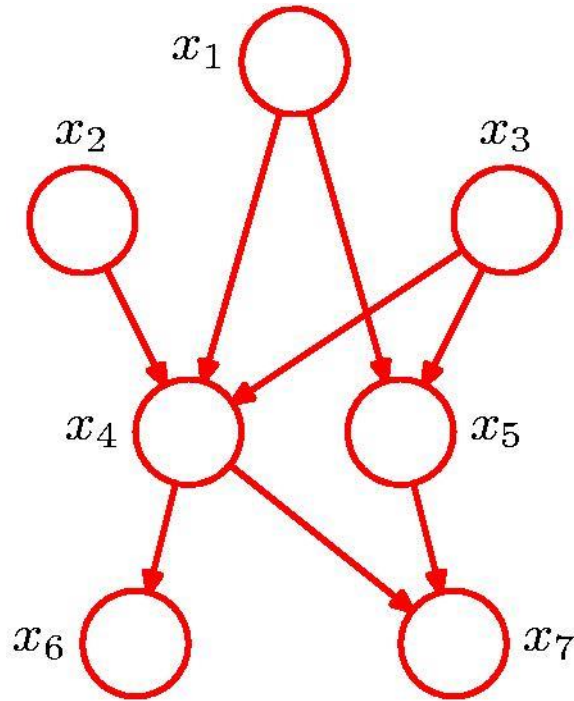
What is $p(a, b, c)$?



$$p(a, b, c) = p(c|a, b)p(a, b) = p(c|a, b)p(b|a)p(a)$$

$$p(x_1, \dots, x_K) = p(x_K|x_1, \dots, x_{K-1}) \dots p(x_2|x_1)p(x_1)$$

Directed graphical models

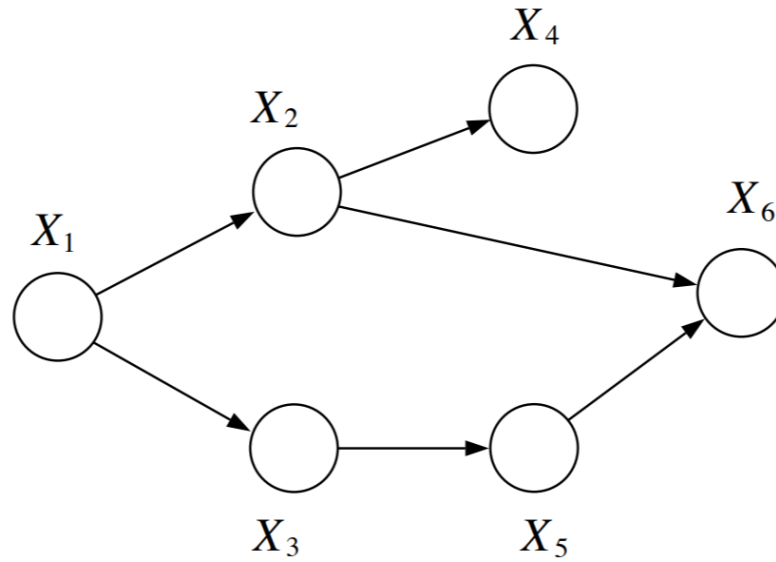


$$p(x_1, \dots, x_7) = p(x_1)p(x_2)p(x_3)p(x_4|x_1, x_2, x_3) \\ p(x_5|x_1, x_3)p(x_6|x_4)p(x_7|x_4, x_5)$$

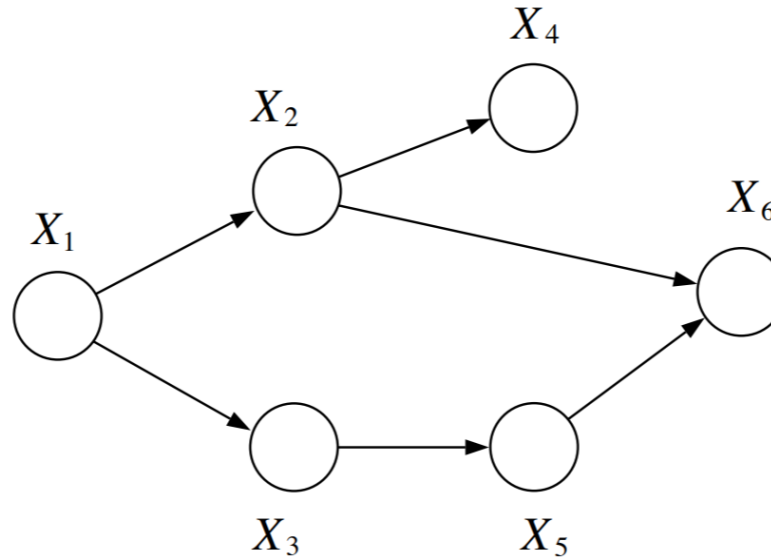
General Factorization

$$p(x) = \prod_{k=1}^K p(x_k | x_{\pi_k})$$

What is $p(X_1, \dots, X_6)$?



What is $p(X_1, \dots, X_6)$?

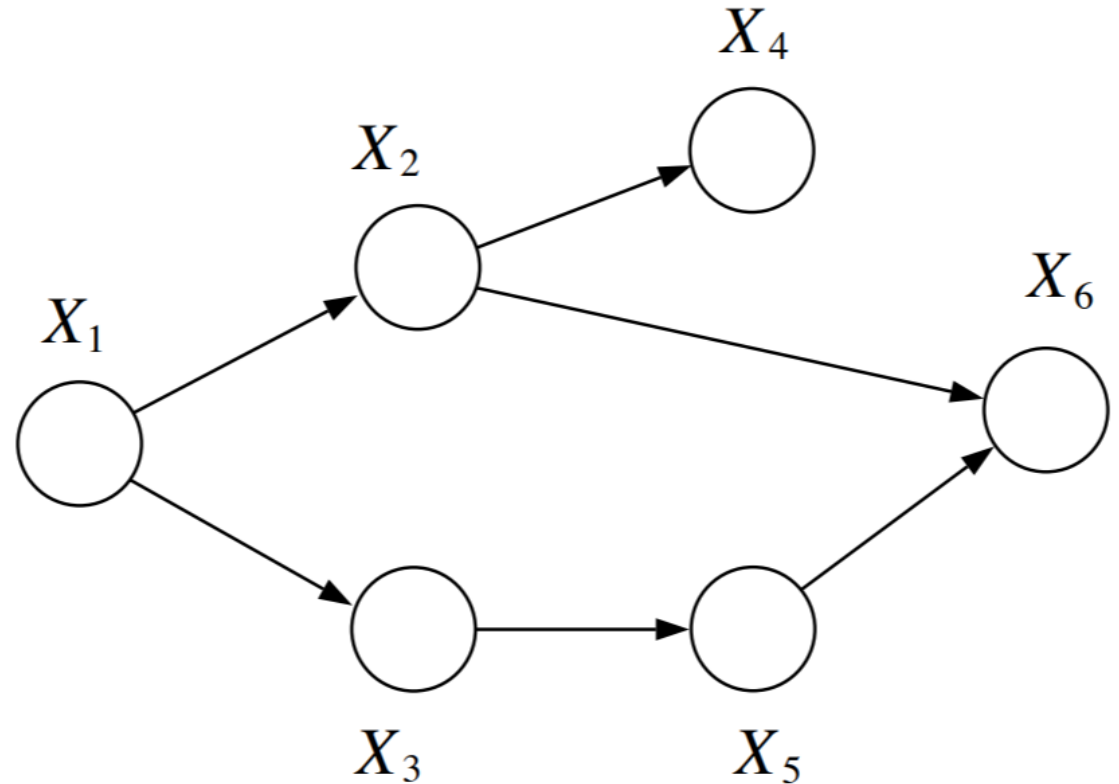


$$p(x_1, x_2, x_3, x_4, x_5, x_6) = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1)p(x_4 \mid x_2)p(x_5 \mid x_3)p(x_6 \mid x_2, x_5),$$

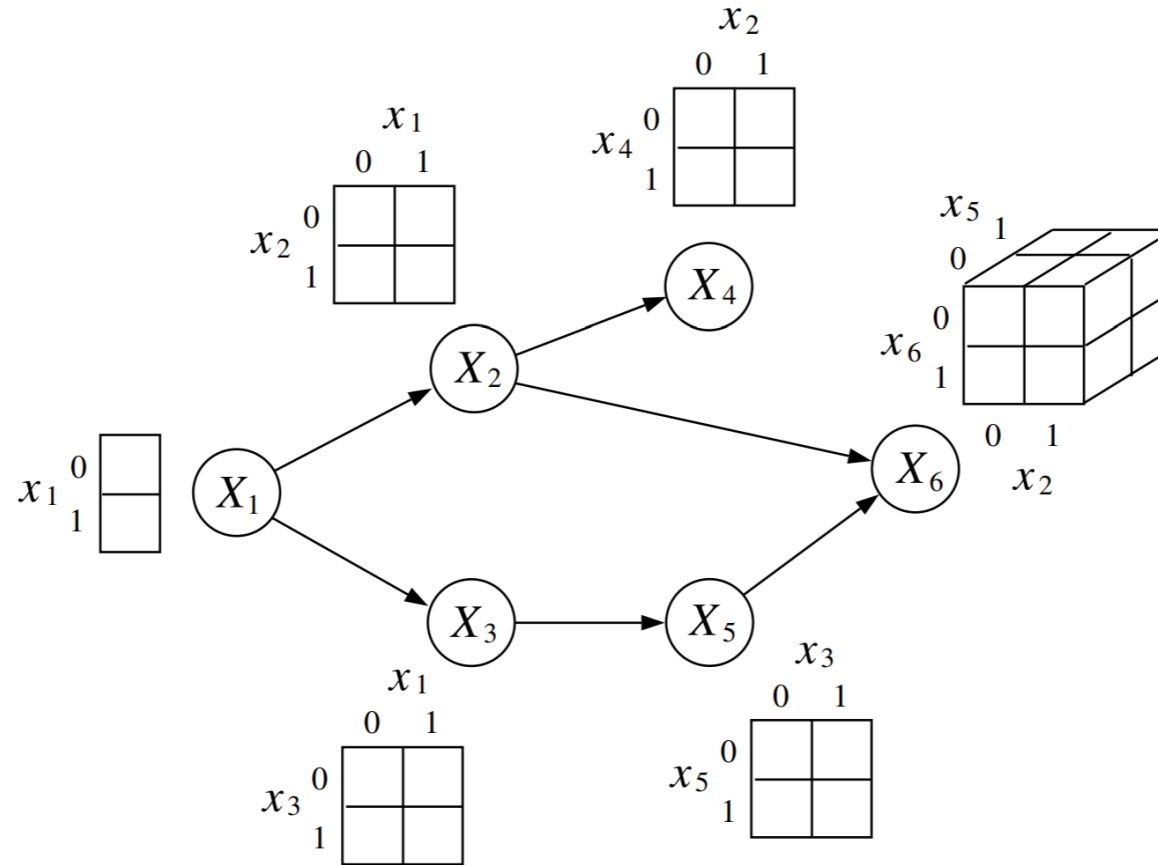
Benefits – Representation

Assume we have n discrete variables each with r outcomes.

We can store only the local probability tables instead of the r^n joint table

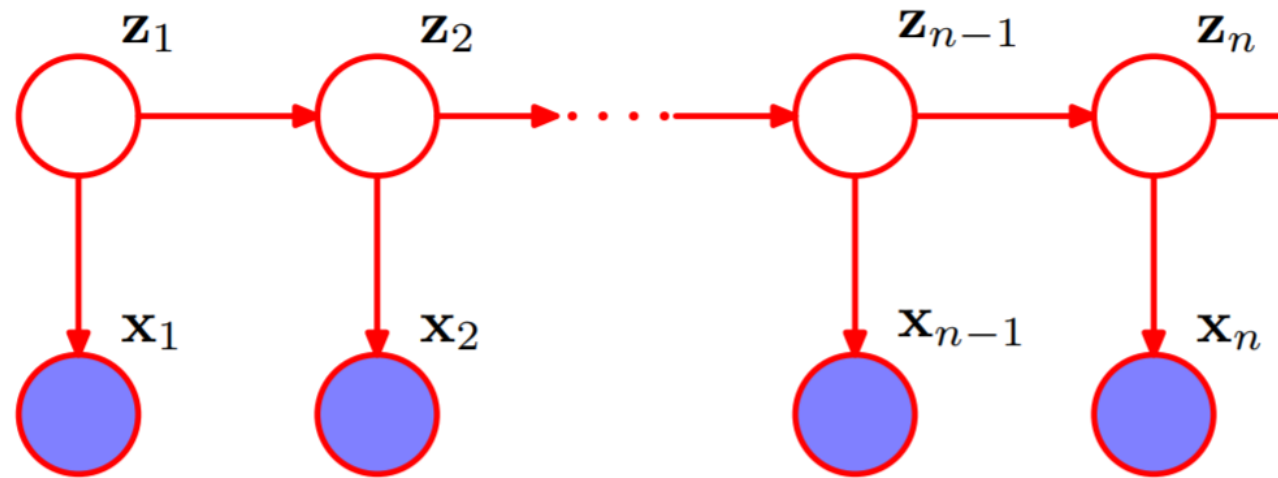


Benefits – Representation



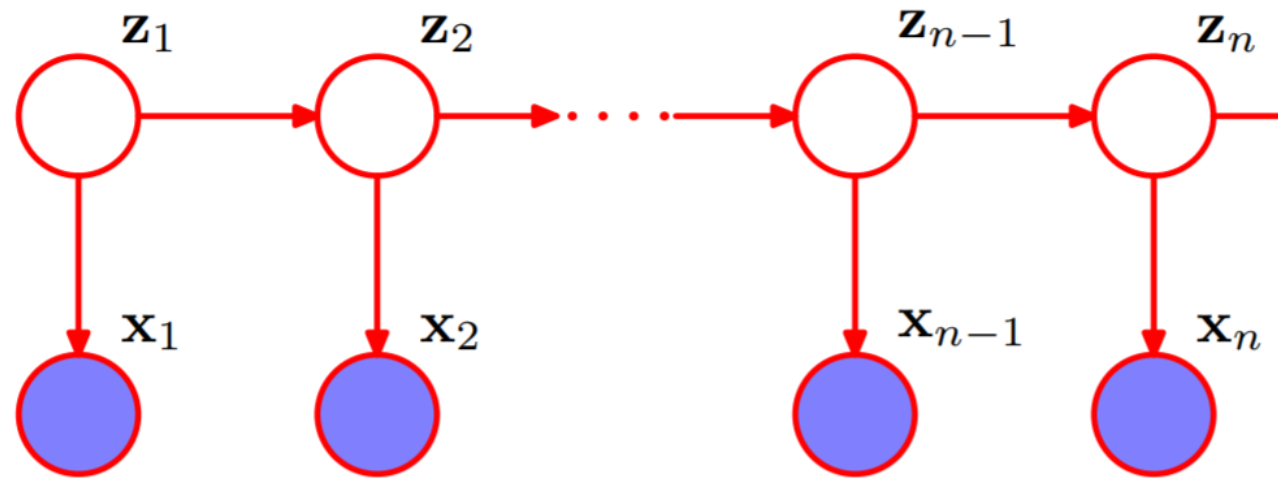
Benefits – Representation

- Hidden Markov Models are a type of model with a highly marginalized structure
- Assume the z variables are binary, what is the complexity computing the joint probability?



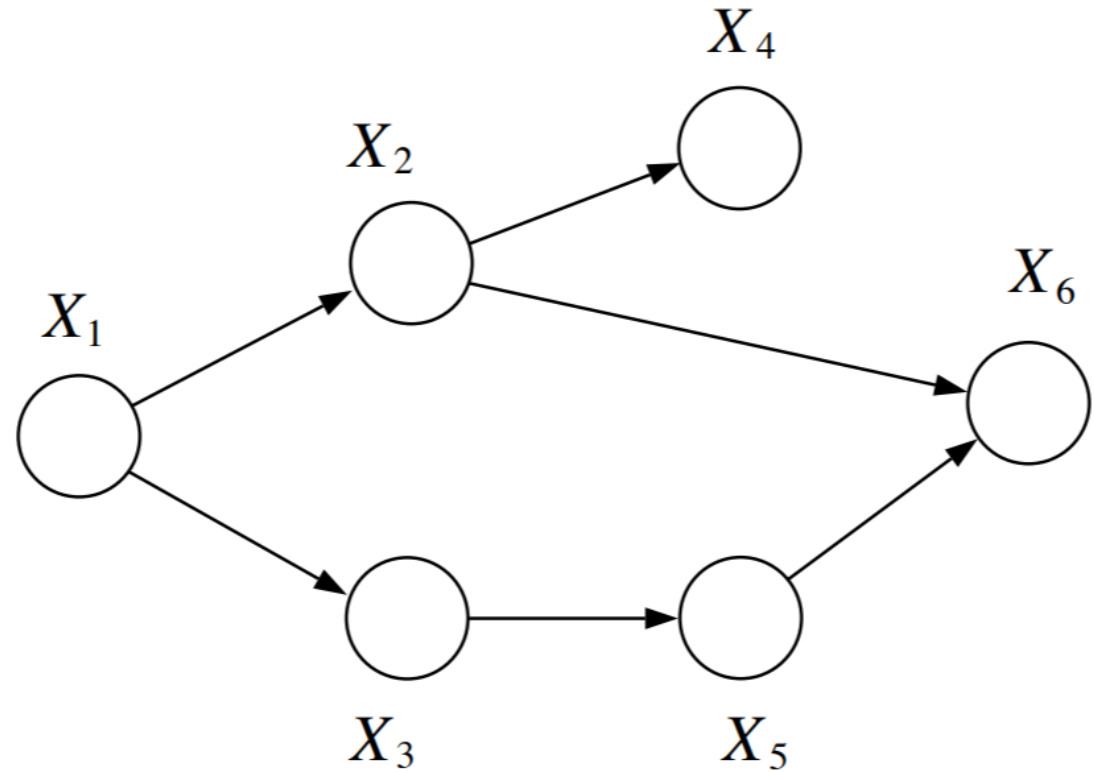
Benefits -- Inference

- In Bayesian ML, data are random variables and we analyze the data by their joint distribution
- The cost of inference is tied to the complexity of the graph



Benefits – Generalizability

- Graphs are more generic than specific joint distributions
- For instance, inference in the graph structure to the right is similar whether the $\{X_1, \dots, X_6\}$ are binary or n -ary
- Or, indeed, discrete and continuous r.v.s



Conditional independence

- We often want to know if X_A and X_B are independent
- We say $X_A \perp\!\!\!\perp X_B$ if $p(x_A, x_B) = p(x_A)p(x_B)$
- Or, if X_A and X_C are conditionally independent given X_B
- Written $X_A \perp\!\!\!\perp X_C \mid X_B$ defined as

$$p(x_A, x_C \mid x_B) = p(x_A \mid x_B)p(x_C \mid x_B)$$

Two factorizations of the joint

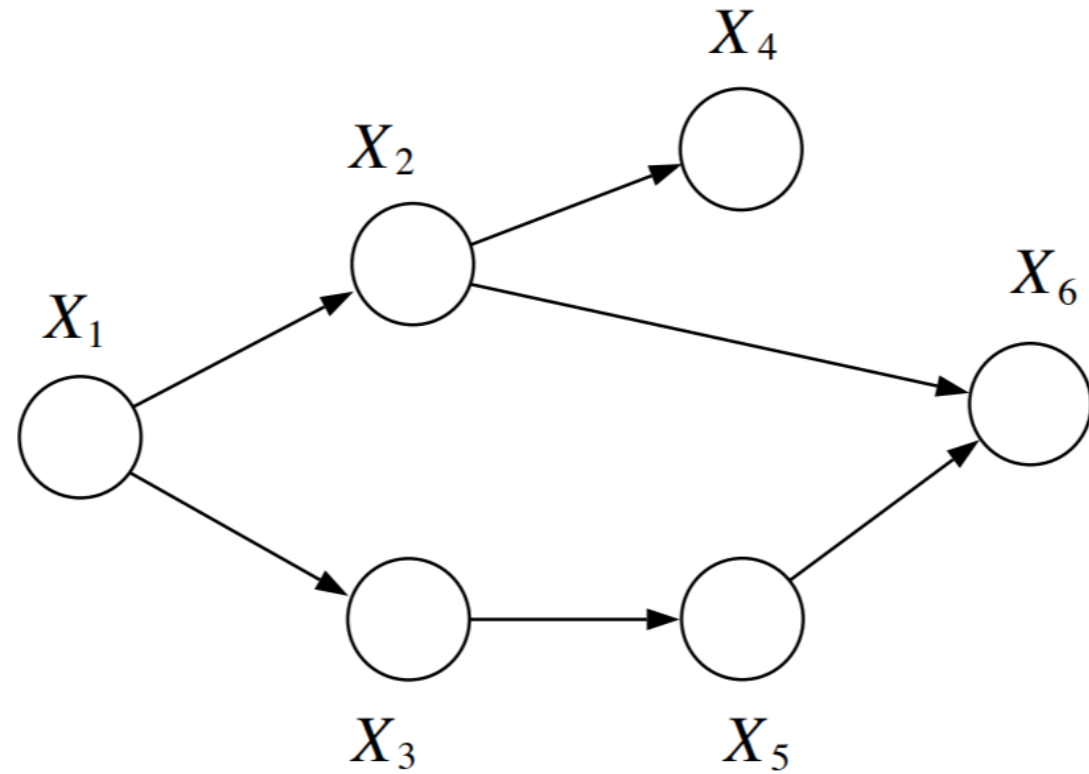
- Chain rule

$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p(x_i \mid x_1, \dots, x_{i-1})$$

- Graphical model

$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p(x_i \mid x_{\pi_i})$$

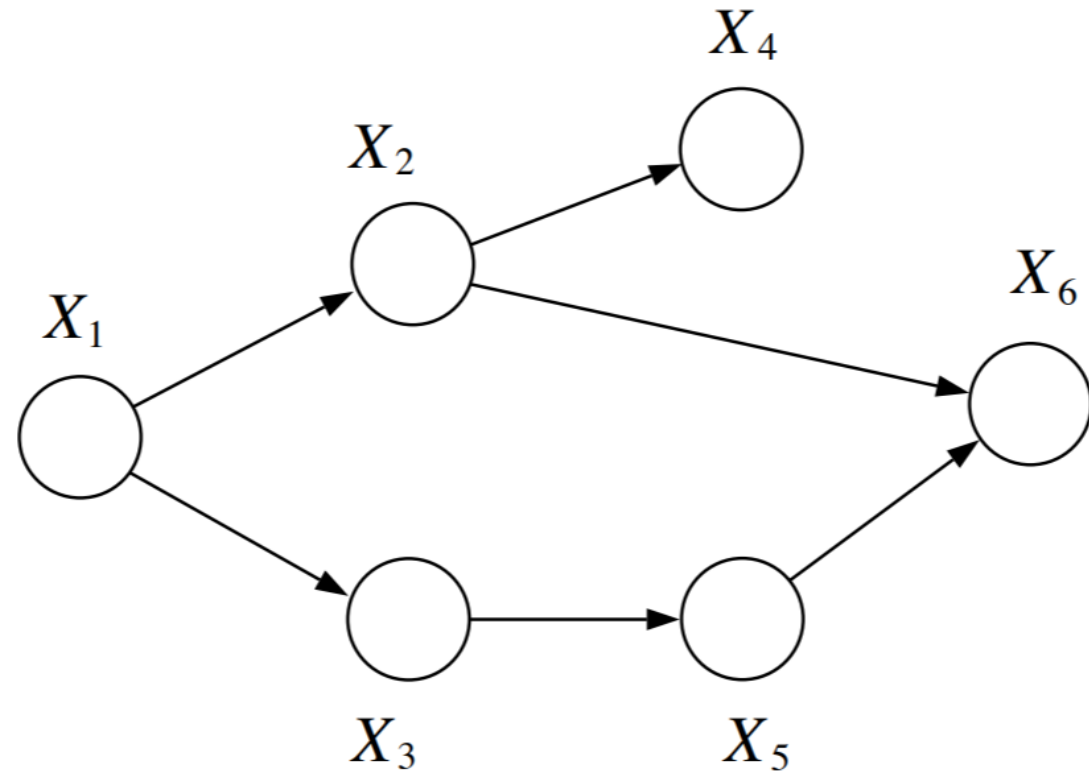
Two factorizations of the joint



Two factorizations of the joint

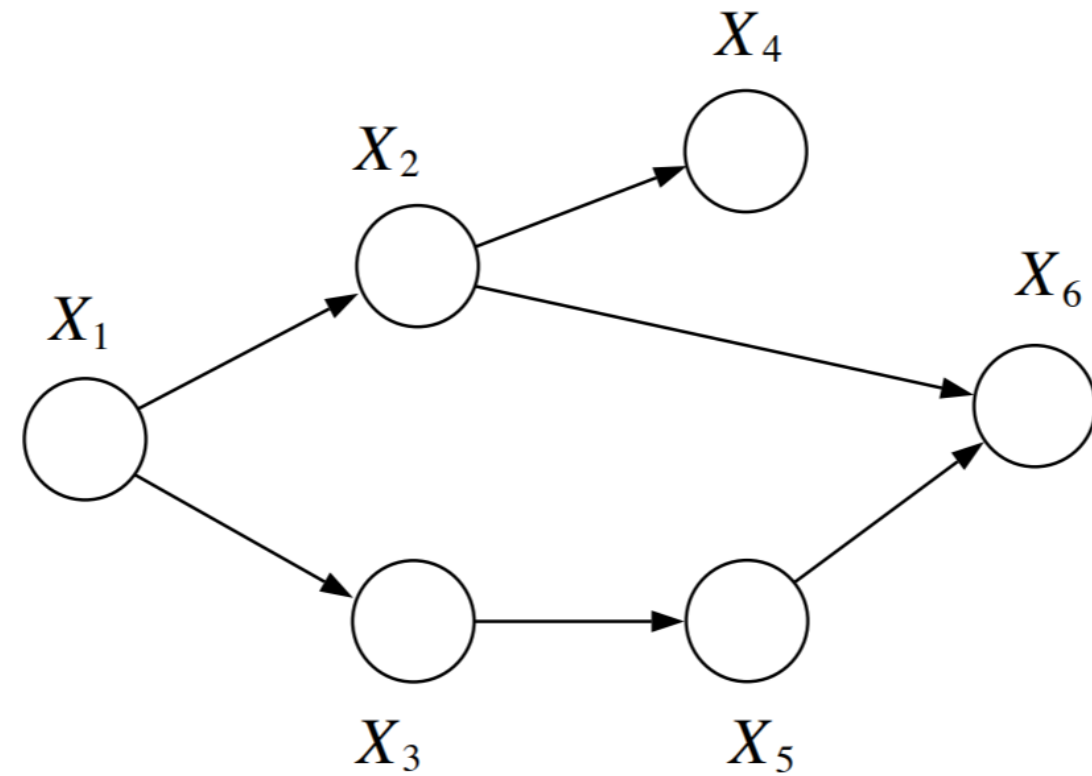
Intuition

missing variables in the conditional probability functions correspond to missing edges in the graphical model



Formalization of intuition

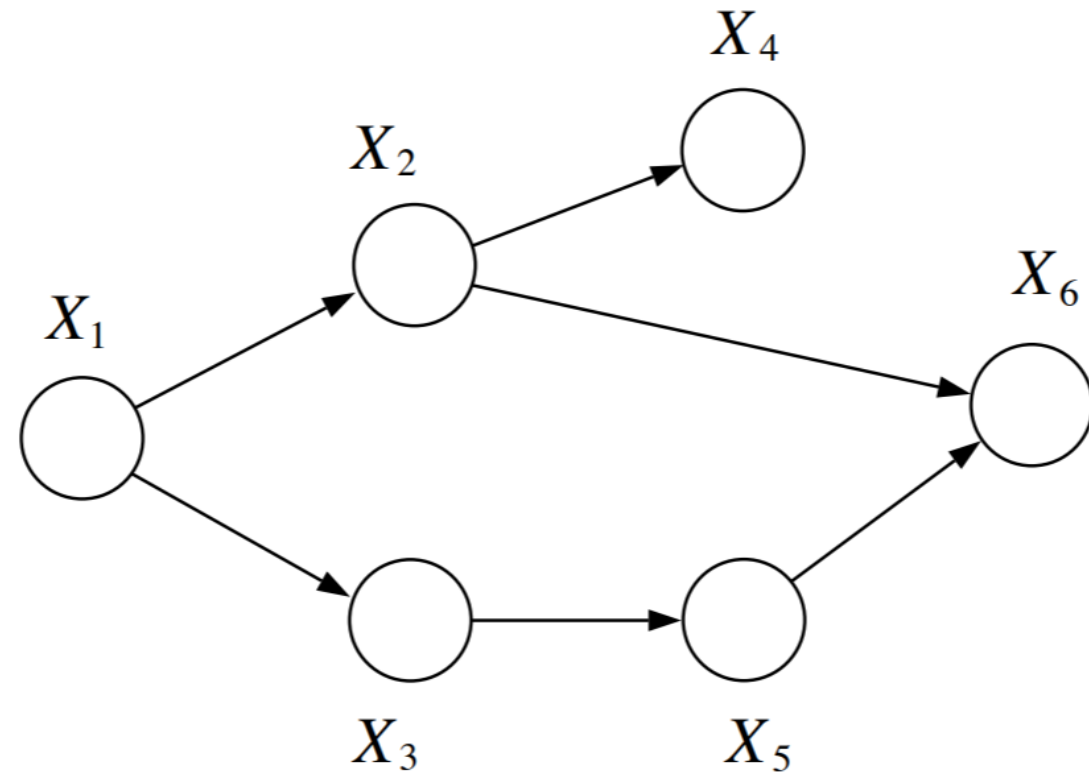
- Define topological ordering I such that nodes in π_i appear before i
- Let v_i denote the nodes that appear earlier than i in our ordering excluding π_i
- v_i contains all non-descendent nodes (including ancestors)



Missing edge conditional independence

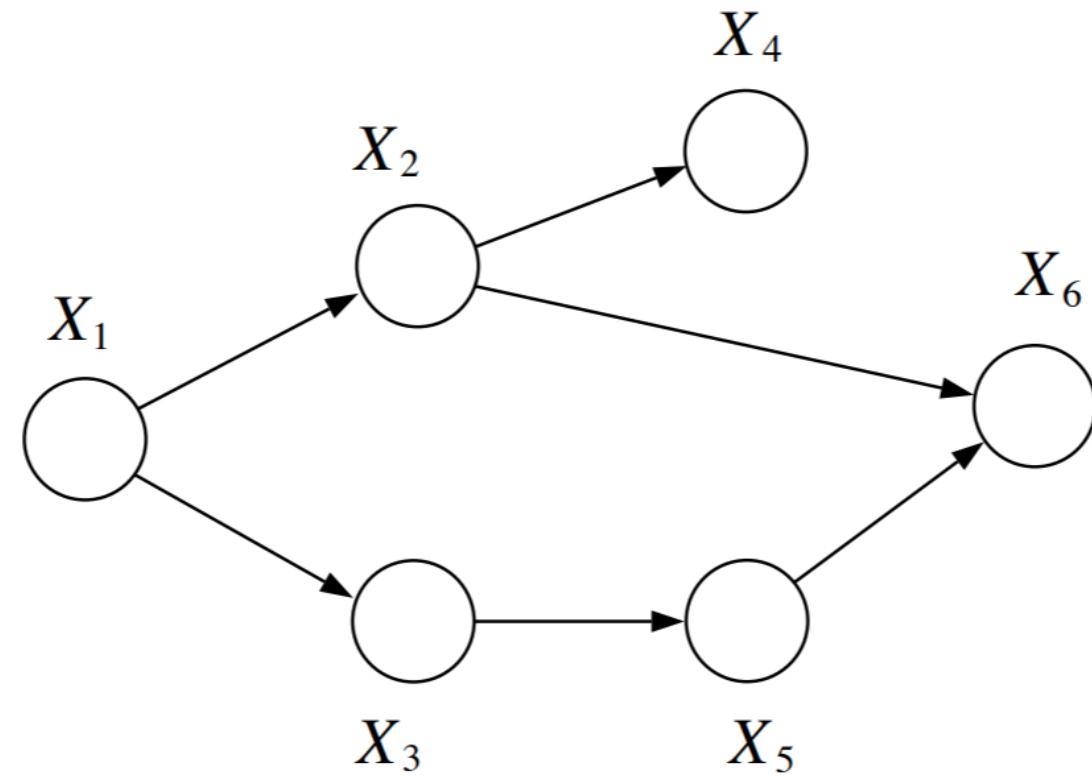
- We associate with the graphical model, a set of basic conditional independence statements

$$\{X_i \perp\!\!\!\perp X_{\nu_i} \mid X_{\pi_i}\}$$



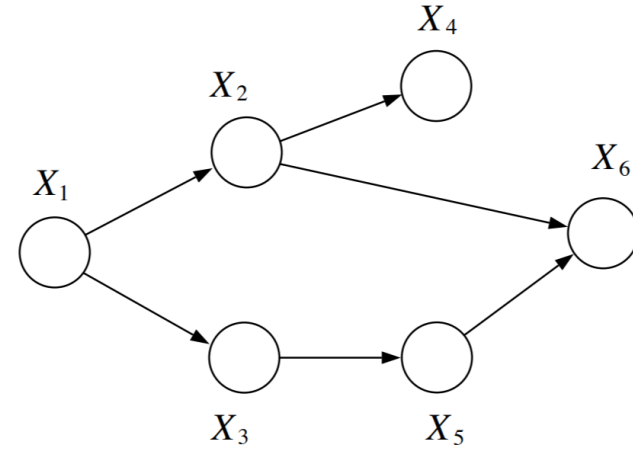
Missing edge conditional independence

$X_1 \perp\!\!\!\perp \emptyset$		$\emptyset$
$X_2 \perp\!\!\!\perp \emptyset$		X_1
$X_3 \perp\!\!\!\perp X_2$		X_1
$X_4 \perp\!\!\!\perp \{X_1, X_3\}$		X_2
$X_5 \perp\!\!\!\perp \{X_1, X_2, X_4\}$		X_3
$X_6 \perp\!\!\!\perp \{X_1, X_3, X_4\}$		$\{X_2, X_5\}$

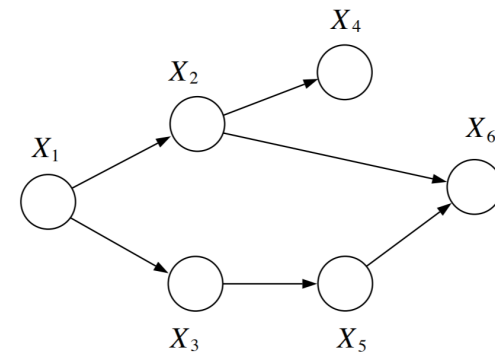


Missing edge conditional independence

- Proof of $X_4 \perp\!\!\!\perp \{X_1, X_3\} \mid X_2$
- $p(x_4|x_1, x_2, x_3) = p(x_4|x_2)$



Proof

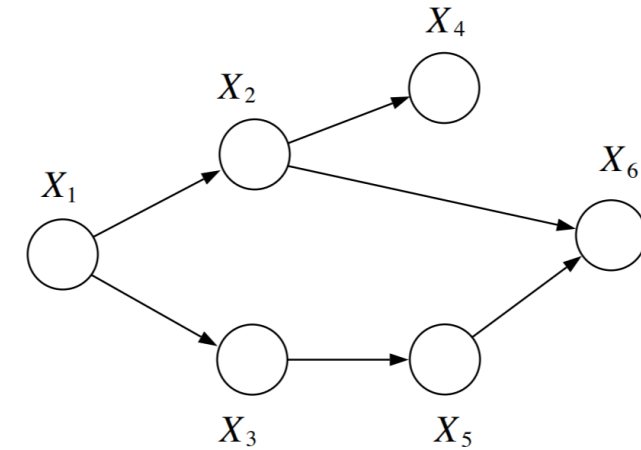


$$\begin{aligned} p(x_1, x_2, x_3, x_4) &= \sum_{x_5} \sum_{x_6} p(x_1, x_2, x_3, x_4, x_5, x_6) \\ &= \sum_{x_5} \sum_{x_6} p(x_1) p(x_2 \mid x_1) p(x_3 \mid x_1) p(x_4 \mid x_2) p(x_5 \mid x_3) p(x_6 \mid x_2, x_5) \\ &= p(x_1) p(x_2 \mid x_1) p(x_3 \mid x_1) p(x_4 \mid x_2) \sum_{x_5} p(x_5 \mid x_3) \sum_{x_6} p(x_6 \mid x_2, x_5) \\ &= p(x_1) p(x_2 \mid x_1) p(x_3 \mid x_1) p(x_4 \mid x_2), \end{aligned}$$

$$\begin{aligned} p(x_1, x_2, x_3) &= \sum_{x_4} p(x_1) p(x_2 \mid x_1) p(x_3 \mid x_1) p(x_4 \mid x_2) \\ &= p(x_1) p(x_2 \mid x_1) p(x_3 \mid x_1). \end{aligned}$$

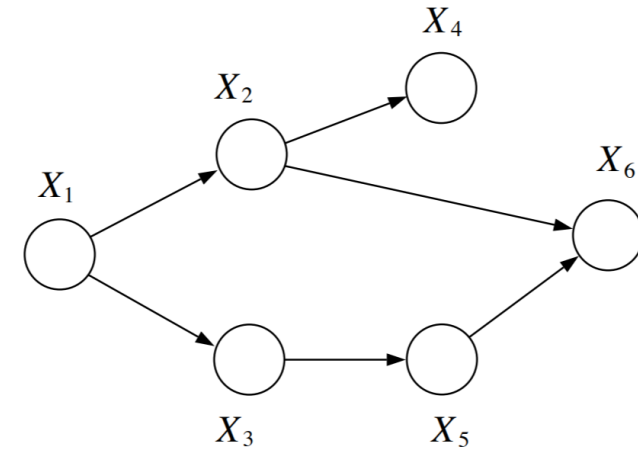
Additional conditional independencies

- Do other conditional independencies follow from the graph structure?

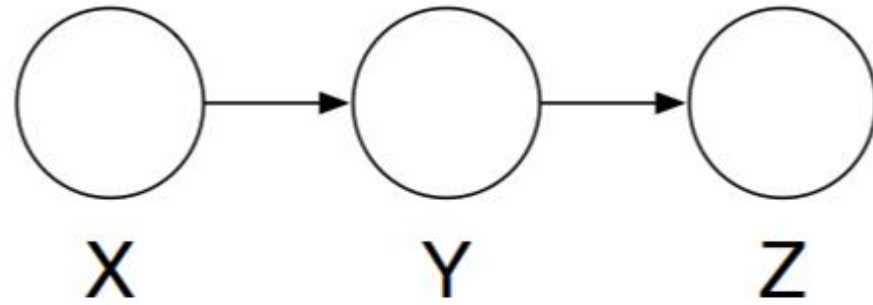


Additional conditional independencies

- Do other conditional independencies follow from the graph structure?
 - Yes – e.g. $X_6 \perp\!\!\!\perp X_1 \mid X_2, X_3$
- Observation:
A node's parents separate it from its ancestors
- So, conditional independence might have something to do with graph separation

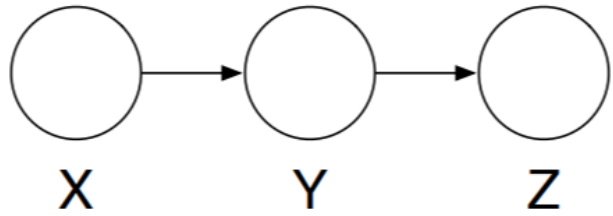


Graph separation – Graph 1



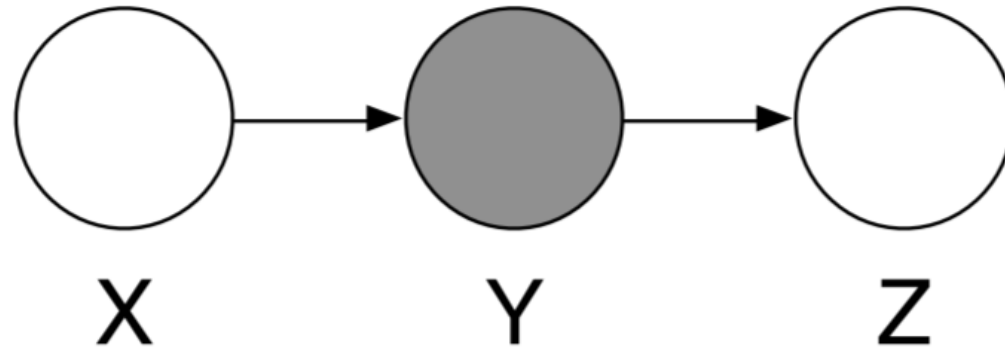
$$p(x, y, z) = p(x)p(y | x)p(z | y).$$

Graph separation – Graph 1

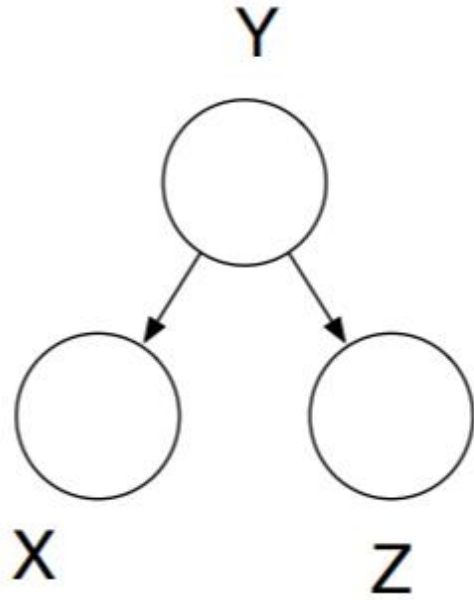


Markov Interpretation

We interpret graph separation as conditional independence



Graph separation – Graph 2



$$p(x, y, z) = p(y)p(x | y)p(z | y)$$

$$X \perp\!\!\!\perp Z \mid Y$$

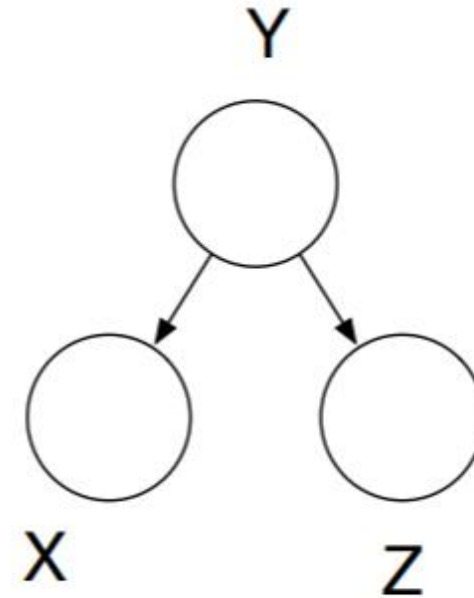
$$\begin{aligned} p(x, z | y) &= \frac{p(y)p(x | y)p(z | y)}{p(y)} \\ &= p(x | y)p(z | y) \end{aligned}$$

Graph separation – Graph 2 Interpretation

$X = \text{shoe size}$

$Z = \text{amount of gray hair}$

- Are they independent?
- What could we condition on to render X and Z conditionally independent?



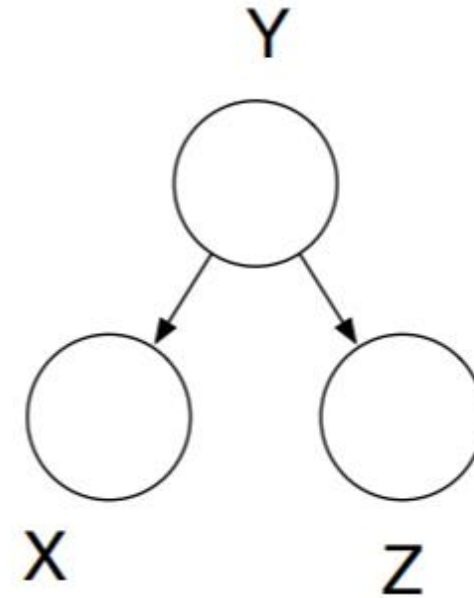
Graph separation – Graph 2 Interpretation

$X = \text{shoe size}$

$Z = \text{amount of gray hair}$

Generally these variables are dependent

But, if $Y = \text{age}$, we can imagine that conditioned on Y , X is independent of Z



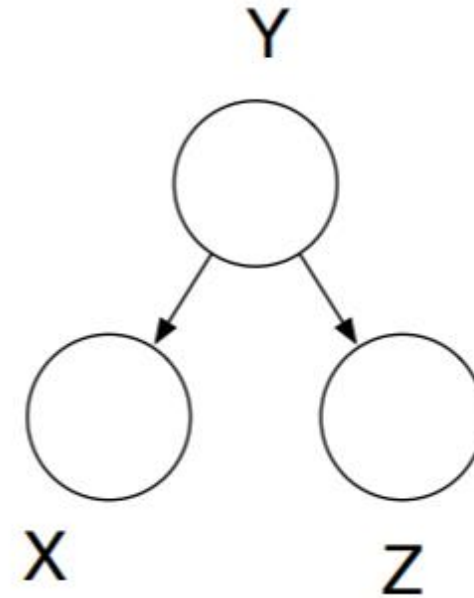
$$p(x, y, z) = p(y)p(x | y)p(z | y)$$

Graph separation – Graph 2 Interpretation

$X = \text{coin flip 1}$
 $Z = \text{coin flip 2}$

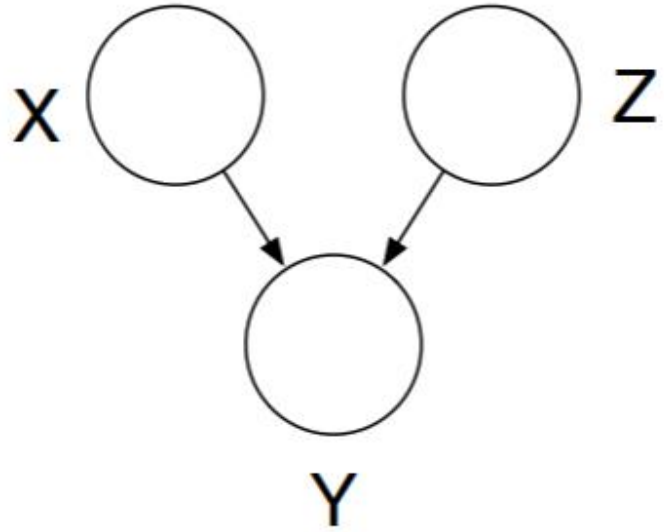
From the same coin, these
variables are not independent

But, if we know the probability
of heads, the flips are
independent



$$p(x, y, z) = p(y)p(x | y)p(z | y)$$

Graph separation – Graph 3

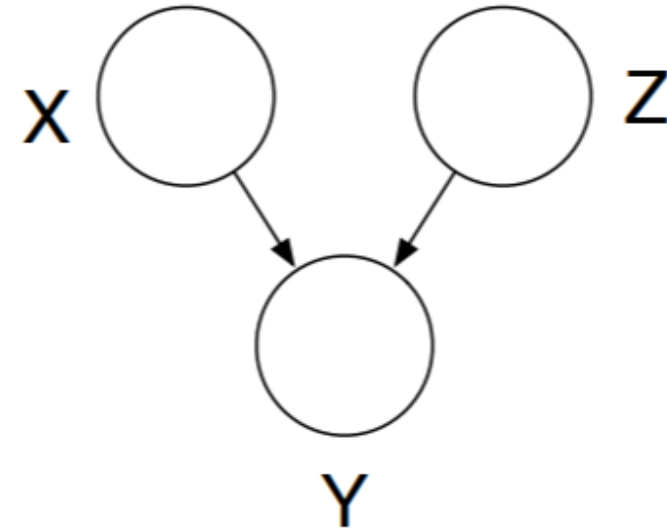


- Independence statement $X \perp\!\!\!\perp Z$
- We can not necessarily say $X \perp\!\!\!\perp Z \mid Y$

$$p(x, y, z) = p(x)p(z)p(y \mid x, z)$$

Graph separation – Graph 3

- Think of causal model:
- $Y = I'm\ late\ for\ lunch$
- $X = I'm\ abducted\ by\ aliens$
- $Z = Car\ breaks\ down$



- Key point
 $P(\text{aliens} = \text{yes} | \text{late} = \text{yes}) \neq P(\text{aliens} = \text{yes} | \text{late} = \text{yes}, \text{broken_car} = \text{no})$

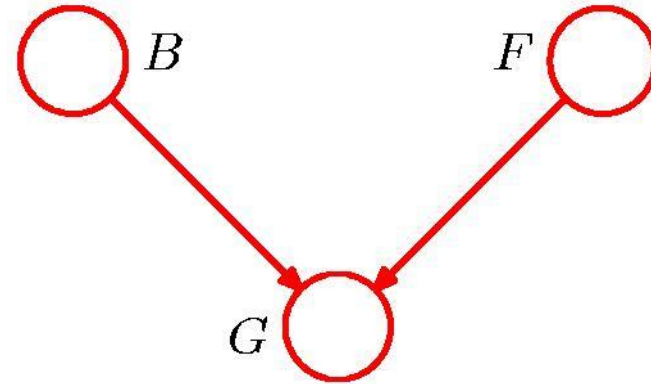
“Am I out of fuel?”

$$p(G = 1|B = 1, F = 1) = 0.8$$

$$p(G = 1|B = 1, F = 0) = 0.2$$

$$p(G = 1|B = 0, F = 1) = 0.2$$

$$p(G = 1|B = 0, F = 0) = 0.1$$



$$p(B = 1) = 0.9$$

$$p(F = 1) = 0.9$$

and hence

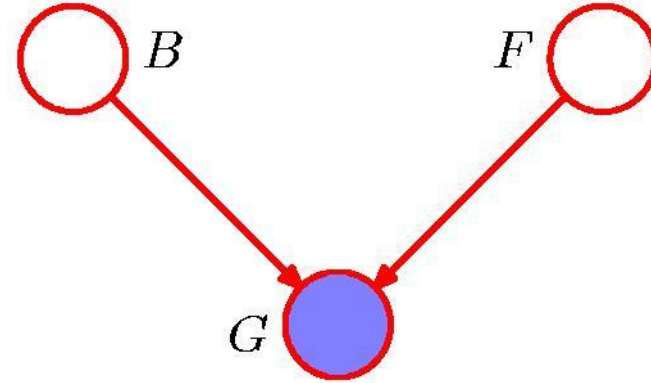
$$p(F = 0) = 0.1$$

B = Battery (0=flat, 1=fully charged)

F = Fuel Tank (0=empty, 1=full)

G = Fuel Gauge Reading
(0=empty, 1=full)

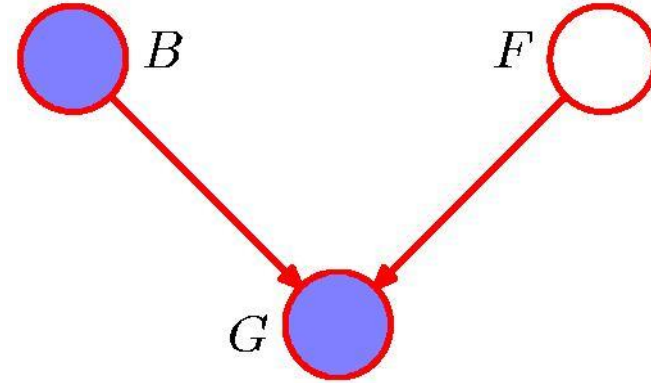
“Am I out of fuel?”



$$\begin{aligned} p(F = 0|G = 0) &= \frac{p(G = 0|F = 0)p(F = 0)}{p(G = 0)} \\ &\simeq 0.257 \end{aligned}$$

Probability of an empty tank increased by observing $G = 0$.

“Am I out of fuel?”



$$\begin{aligned} p(F = 0 | G = 0, B = 0) &= \frac{p(G = 0 | B = 0, F = 0)p(F = 0)}{\sum_{F \in \{0,1\}} p(G = 0 | B = 0, F)p(F)} \\ &\simeq 0.111 \end{aligned}$$

Probability of an empty tank reduced by observing $B = 0$.
This referred to as “explaining away”.

d-separation

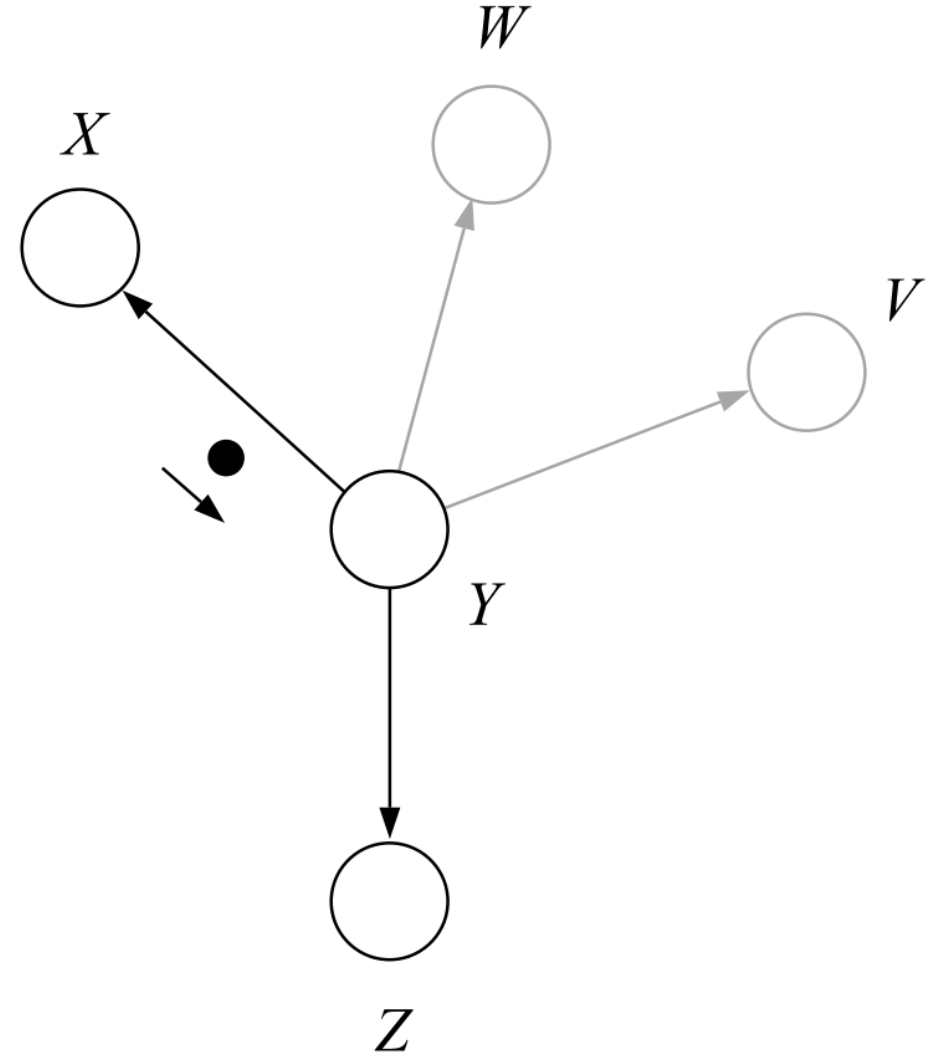
- Goal: determine if a conditional independence statement is valid for a particular graphical model
- Graphs 1 and 2 show situations which naïve graph separation corresponds to conditional independence
- Graph 3 shows a situation where naïve graph separation does not correspond to conditional independence
- Instead of using naïve separation we use d-separation through the Bayes ball algorithm

Bayes ball algorithm

- Bayes ball is a “reachability” algorithm under a particular definition of separation
- Suppose we wish to show $X_A \perp\!\!\!\perp X_B \mid X_C$
- Bayes ball decides whether the conditioned nodes d-separate the nodes on either side of the independence relationship

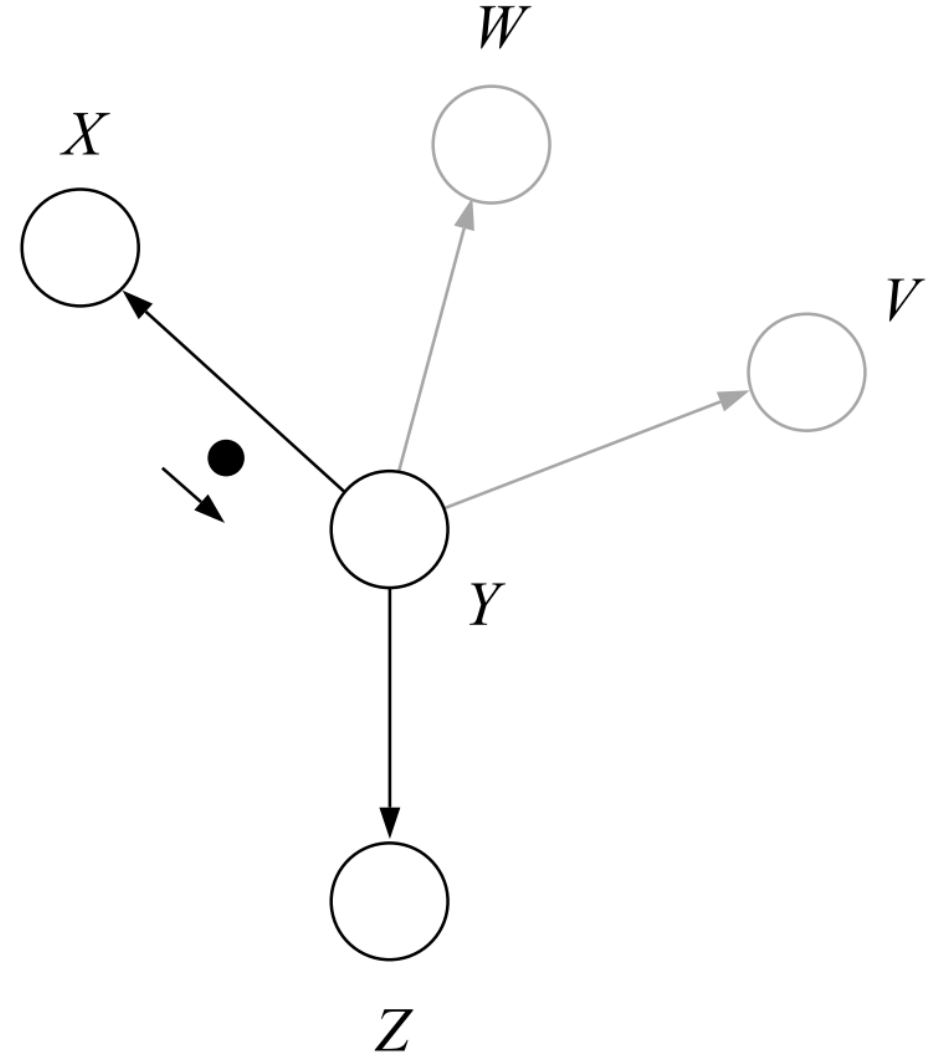
Bayes ball algorithm

- Balls start from sets of variables
 - if they reach each other, conditional independence is false



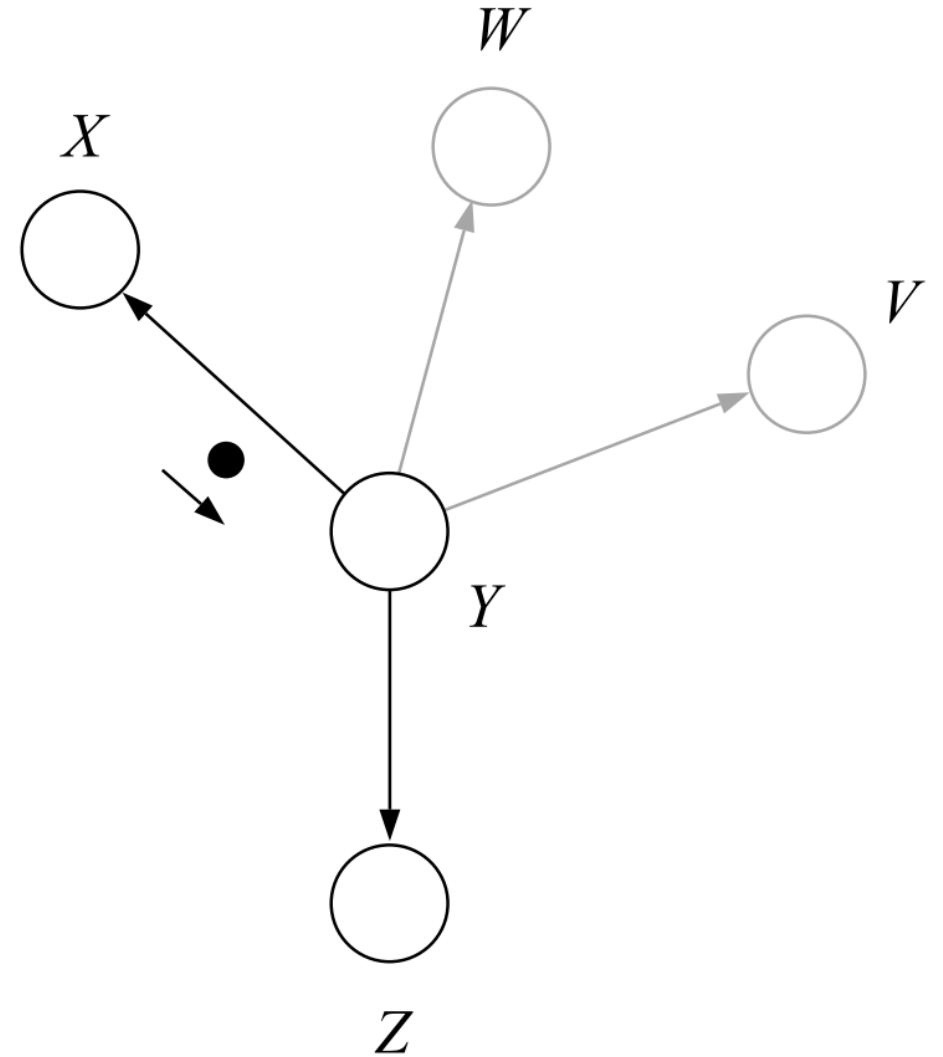
Bayes ball algorithm

- Balls start from sets of variables
 - if they reach each other, conditional independence is false
- Per iteration, a ball starts at X going through Y on its way to Z
 - each ball has a direction and orientation



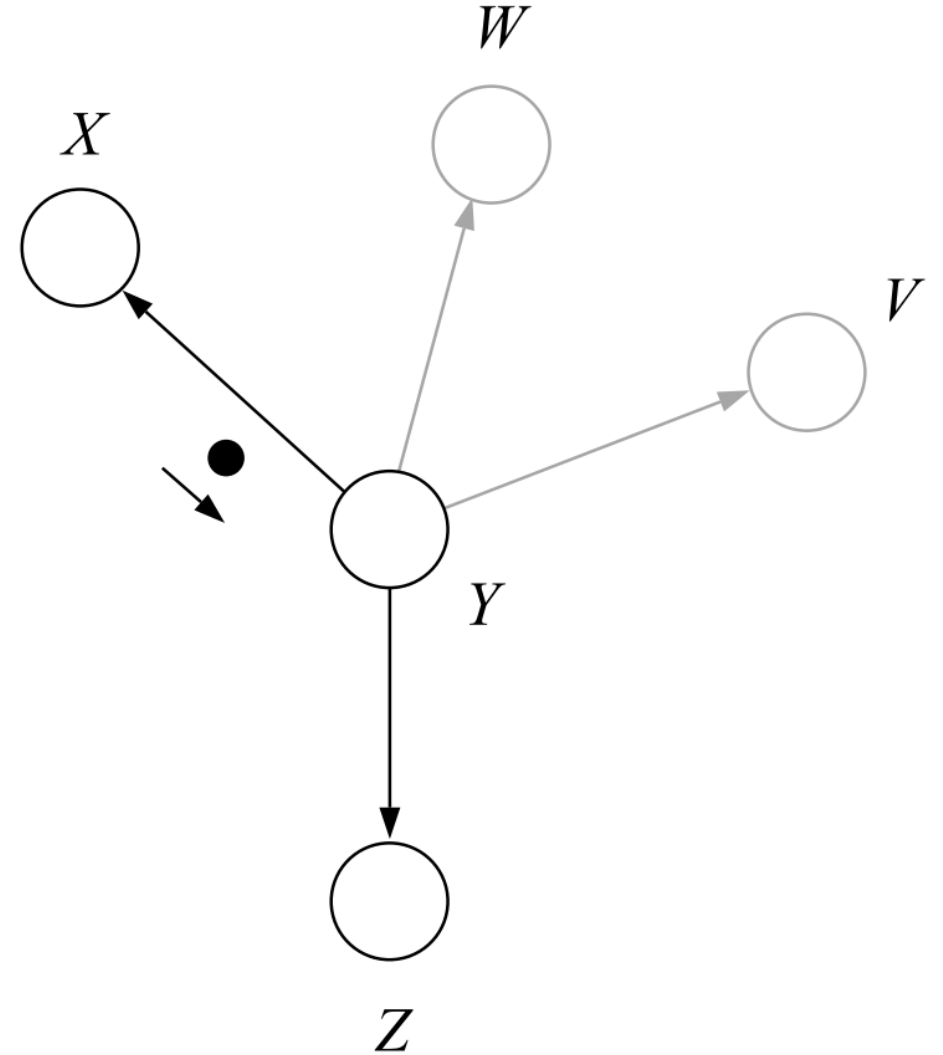
Bayes ball algorithm

- Balls start from sets of variables
 - if they reach each other, conditional independence is false
- Per iteration, a ball starts at X going through Y on its way to Z
 - each ball has a direction and orientation
- If move is allowed
 - ball is at Y and goes through Z to another node

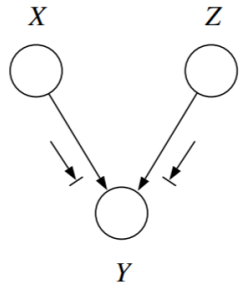
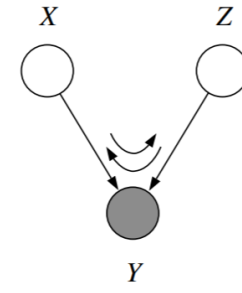
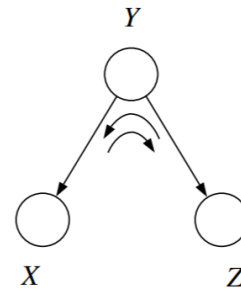
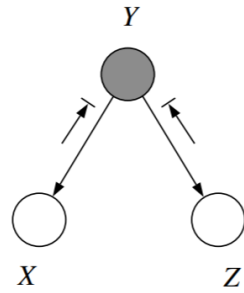
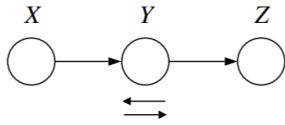
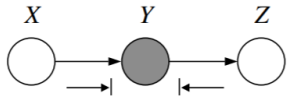


Bayes ball algorithm

- Balls start from sets of variables
 - if they reach each other, conditional independence is false
- Per iteration, a ball starts at X going through Y on its way to Z
 - each ball has a direction and orientation
- If move is allowed
 - ball is at Y and goes through Z to another node
- It does not matter if source X and destination Z are shaded



Rules for conditional independence



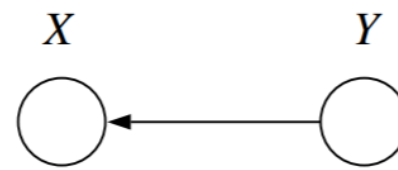
Graph A



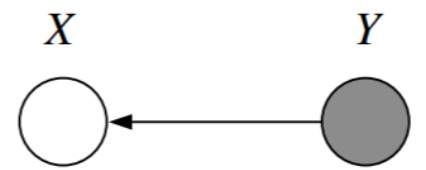
Graph B



Graph C

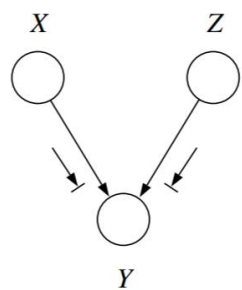
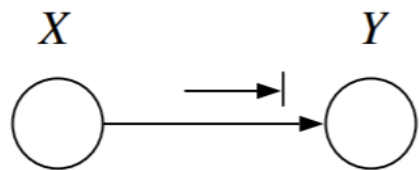


Graph D

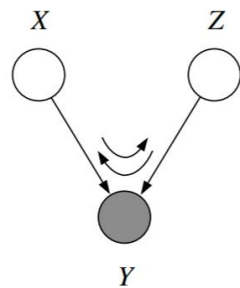
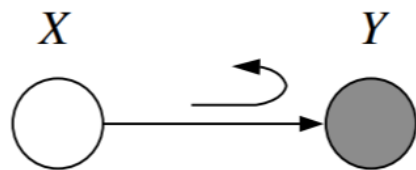


Assume $X=Z$, which of these paths are blocked?

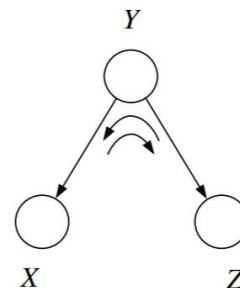
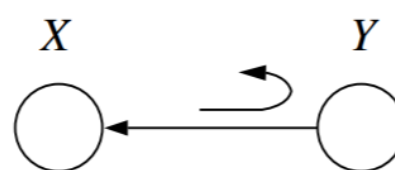
Graph A



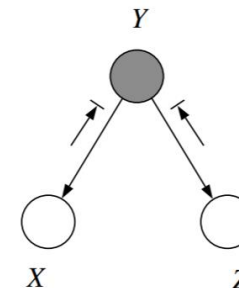
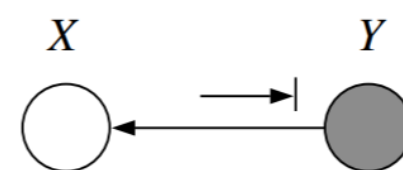
Graph B



Graph C



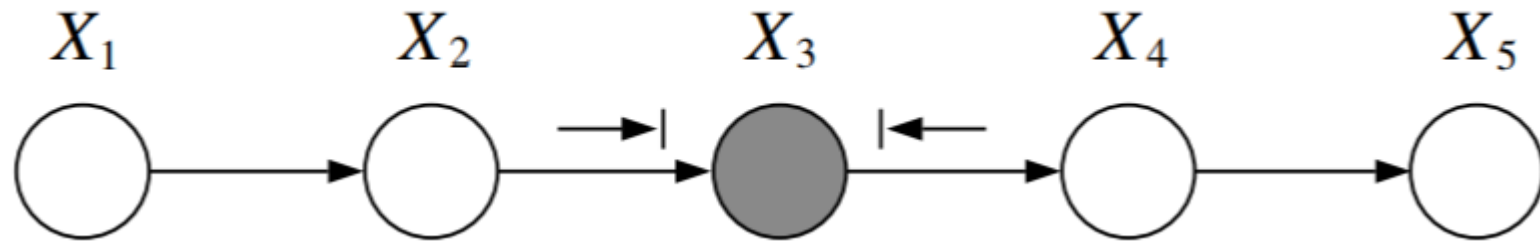
Graph D



Example: Markov chain

- Basic conditional independencies are

$$X_{i+1} \perp\!\!\!\perp \{X_1, X_2, \dots, X_{i-1}\} \mid X_i.$$



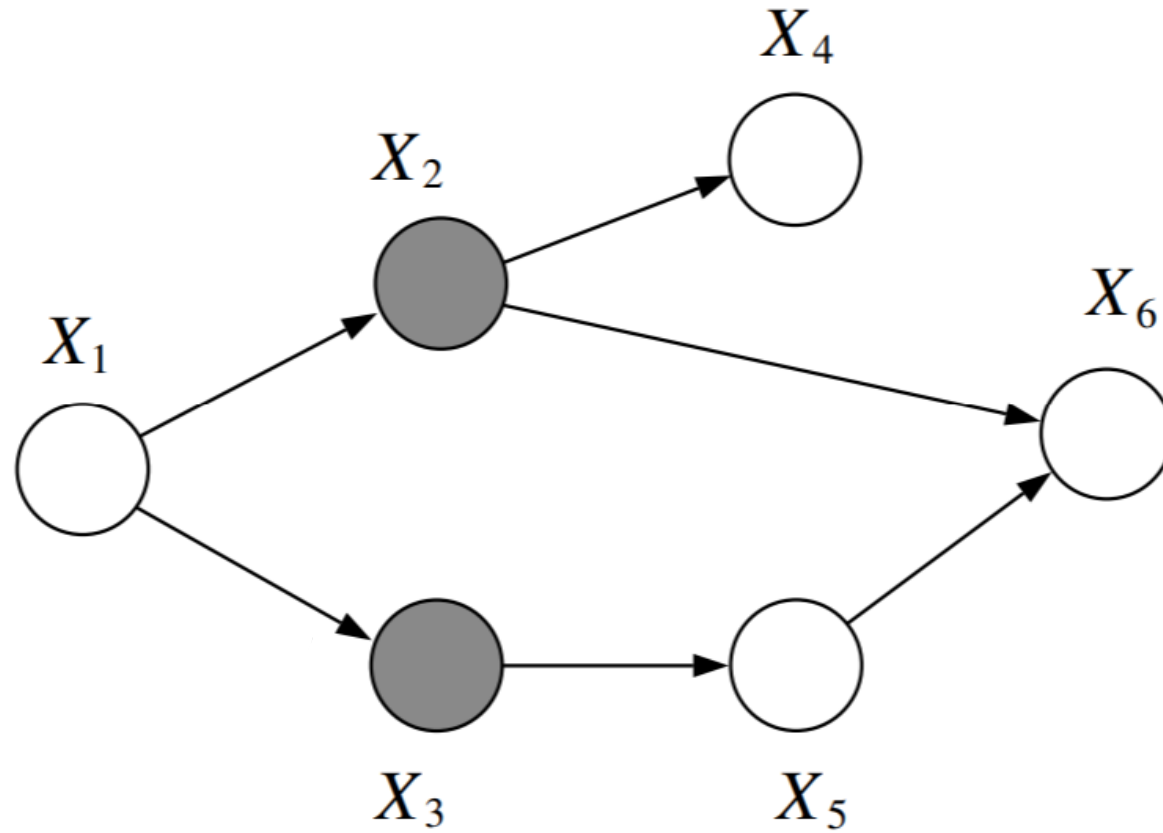
$$X_1 \perp\!\!\!\perp X_5 \mid X_4,$$

$$X_1 \perp\!\!\!\perp X_5 \mid X_2,$$

$$X_1 \perp\!\!\!\perp X_5 \mid \{X_2, X_4\},$$

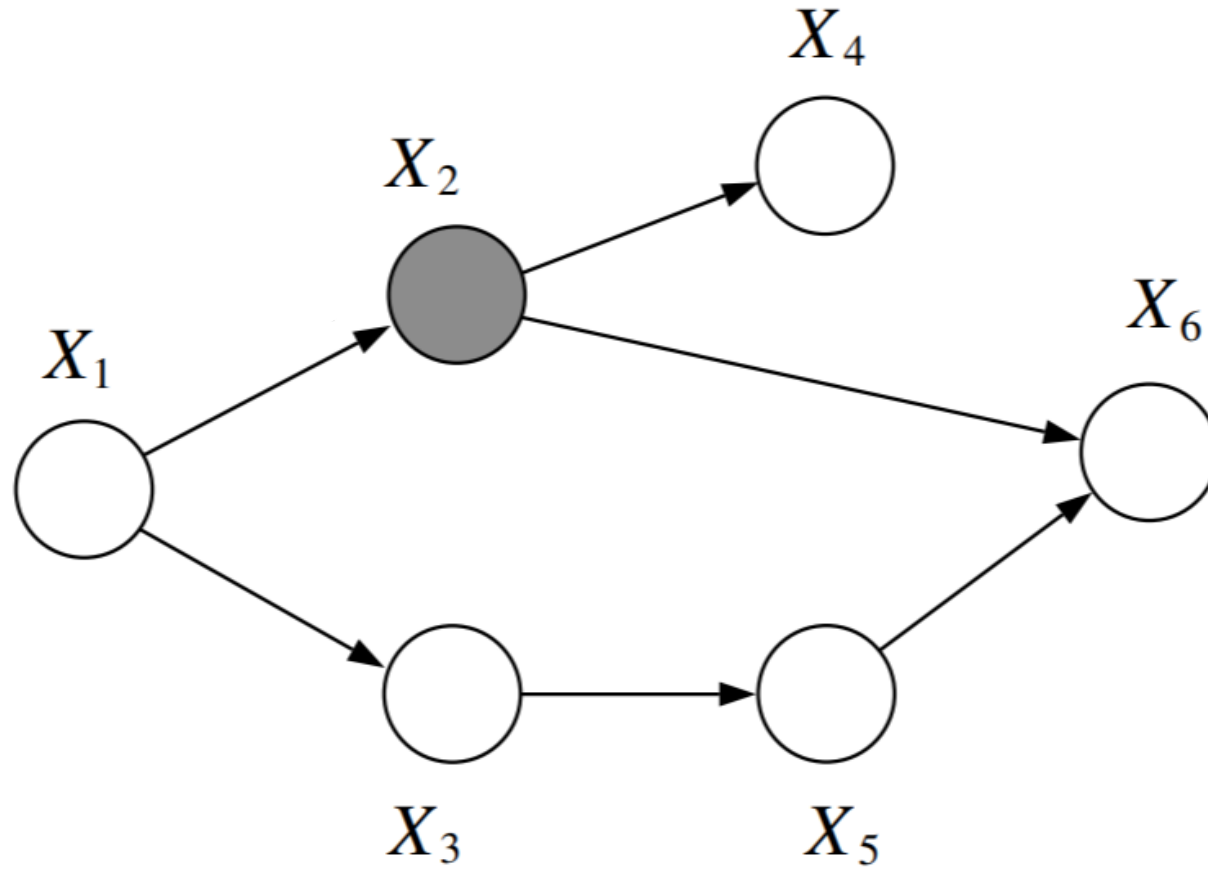
Example: general graph

$X_4 \perp\!\!\!\perp X_6 \mid \{X_2, X_3\} ?$



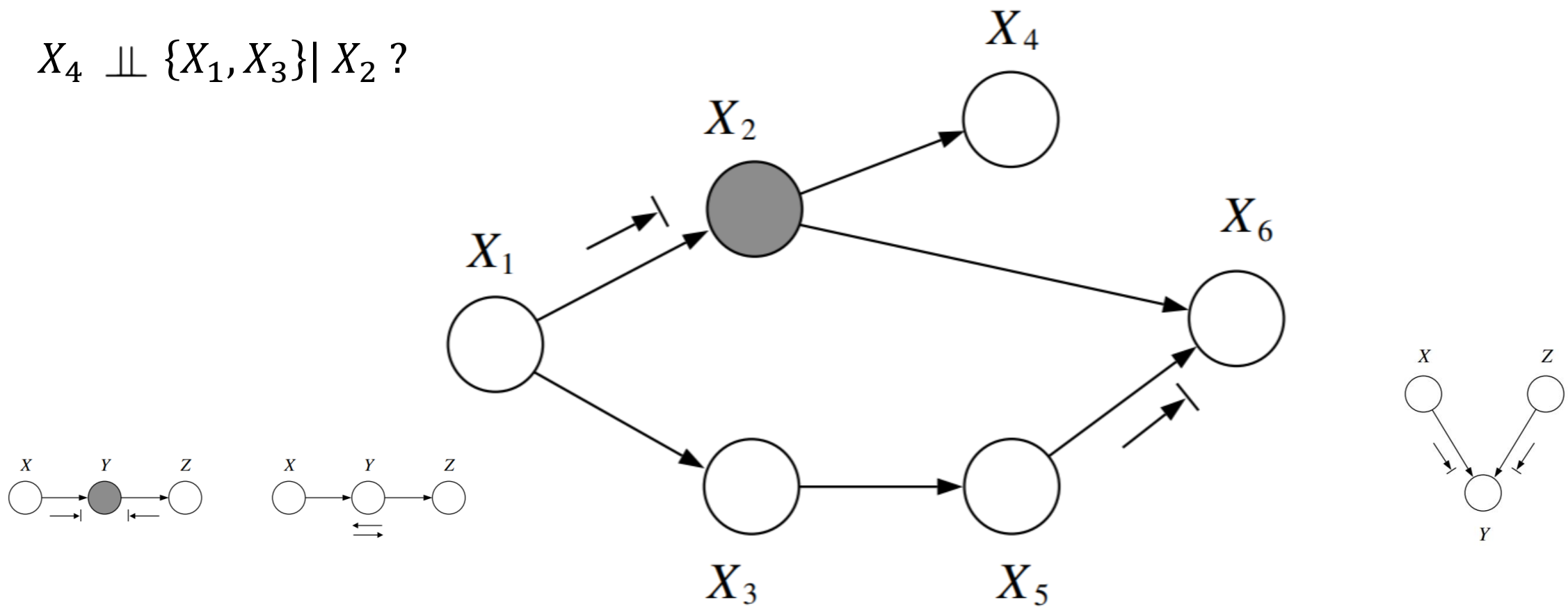
Example: general graph

$X_4 \perp\!\!\!\perp \{X_1, X_3\} \mid X_2$?



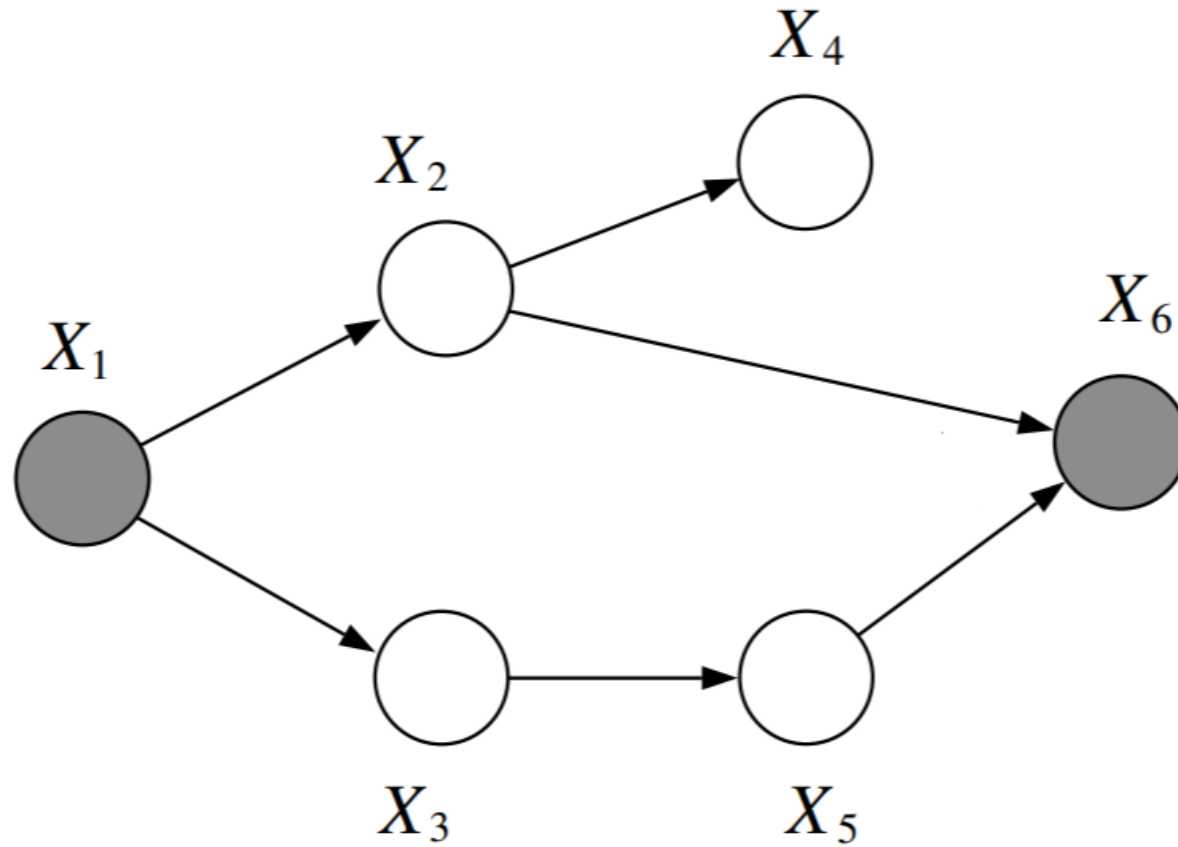
Example: general graph

$X_4 \perp\!\!\!\perp \{X_1, X_3\} \mid X_2 \text{ ?}$



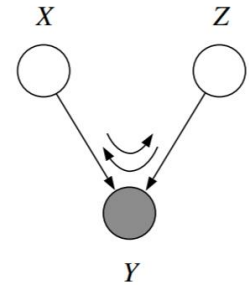
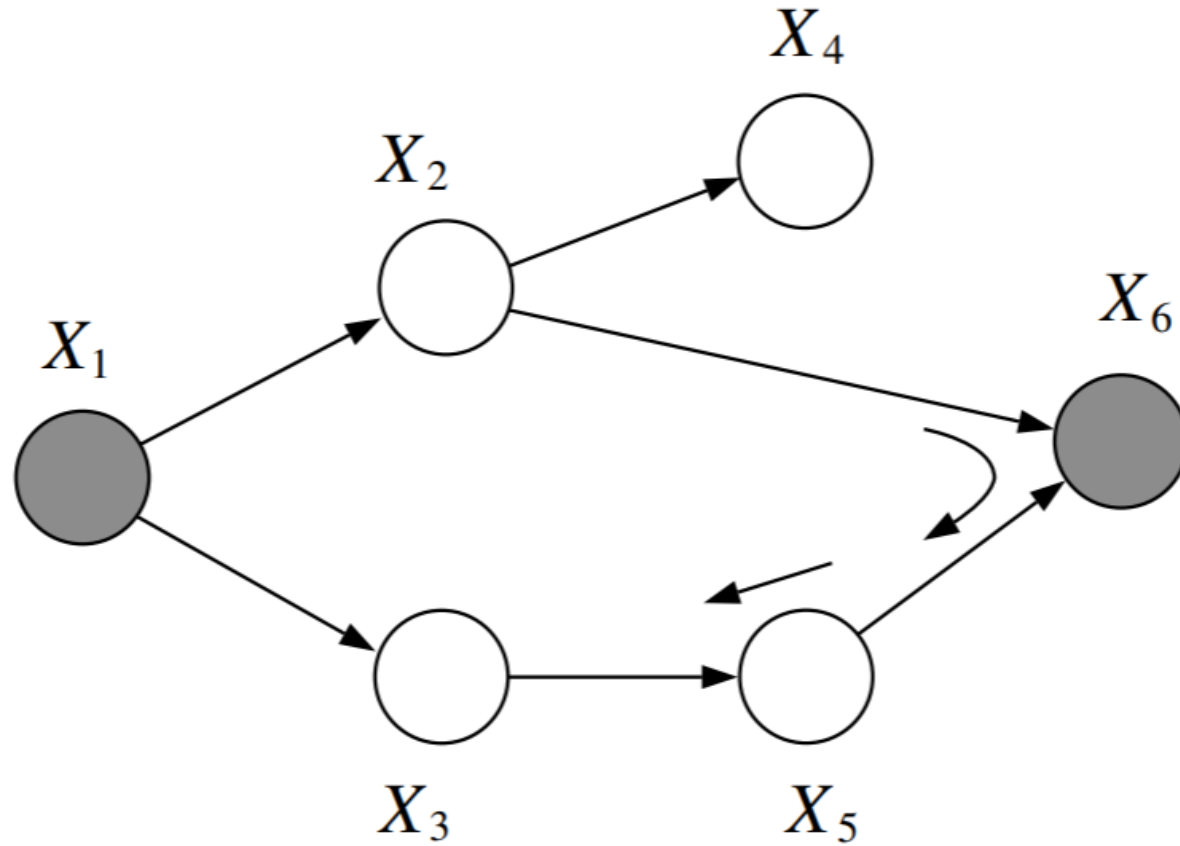
Example general graph

$X_2 \perp\!\!\!\perp X_3 \mid \{X_1, X_6\} ?$

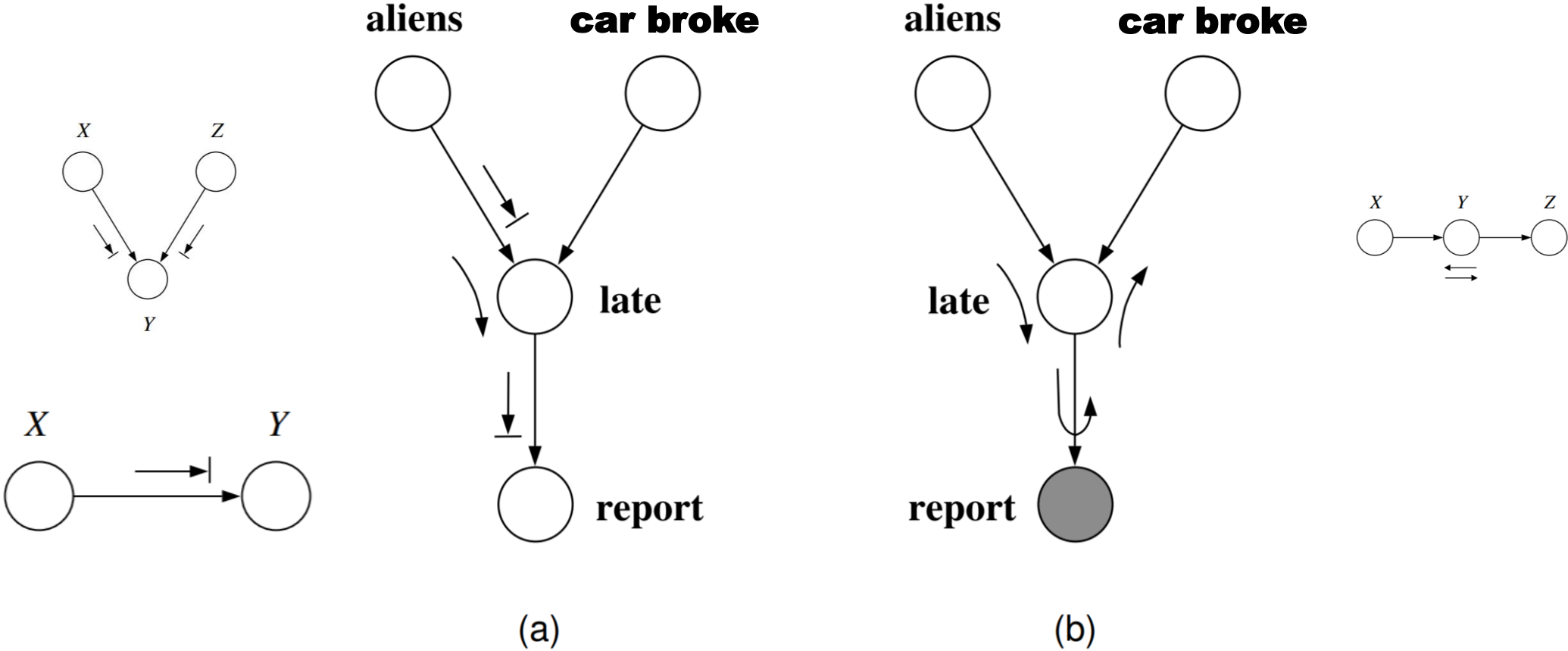


Example general graph

$$X_2 \perp\!\!\!\perp X_3 \mid \{X_1, X_6\} ?$$



Example general graph



Hammersley-Clifford theorem

- Consider two probability distributions obtained from graphical model G

family of joints found by ranging
over all conditional probability
tables associated with G

$$p(x) = \prod_{k=1}^K p(x_k | x_{\pi_k})$$

all joints implied by G and d-separation

1. compute every conditional independence
2. consider every joint distribution on the same set of variables
3. check each joint, if one conditional independence does not hold, throw away joint

- **Hammersley-Clifford proves that these two families are the same**

Example: naïve Bayes Classifier

- Classify an email as spam ($Y = 1$) or not spam ($Y = 0$)
 - Let $i = 1, \dots, n$ index the n words in the English language
 - $X_i = 1$ if word i appears in email, otherwise 0
 - Emails are drawn from $p(Y, X_1, \dots, X_n)$

- We want to compute

$$p(Y = 1 | x_1, \dots, x_n) = \frac{p(x_1, \dots, x_n | Y = 1)p(Y = 1)}{p(x_1, \dots, x_n)}$$

- What quantities are difficult to compute?

Example: naïve Bayes Classifier

$$p(Y = 1 | x_1, \dots, x_n) = \frac{p(x_1, \dots, x_n | Y = 1)p(Y = 1)}{p(x_1, \dots, x_n)}$$

$$p(y, x_1, \dots, x_n) = p(y) \prod_{i=1}^n p(x_i | y)$$

$$p(Y = 1 | x_1, \dots, x_n) = \frac{p(Y = 1) \prod_{i=1}^n p(x_i | Y = 1)}{\sum_{y=\{0,1\}} p(Y = y) \prod_{i=1}^n p(x_i | Y = y)}$$

Example: naïve Bayes Classifier

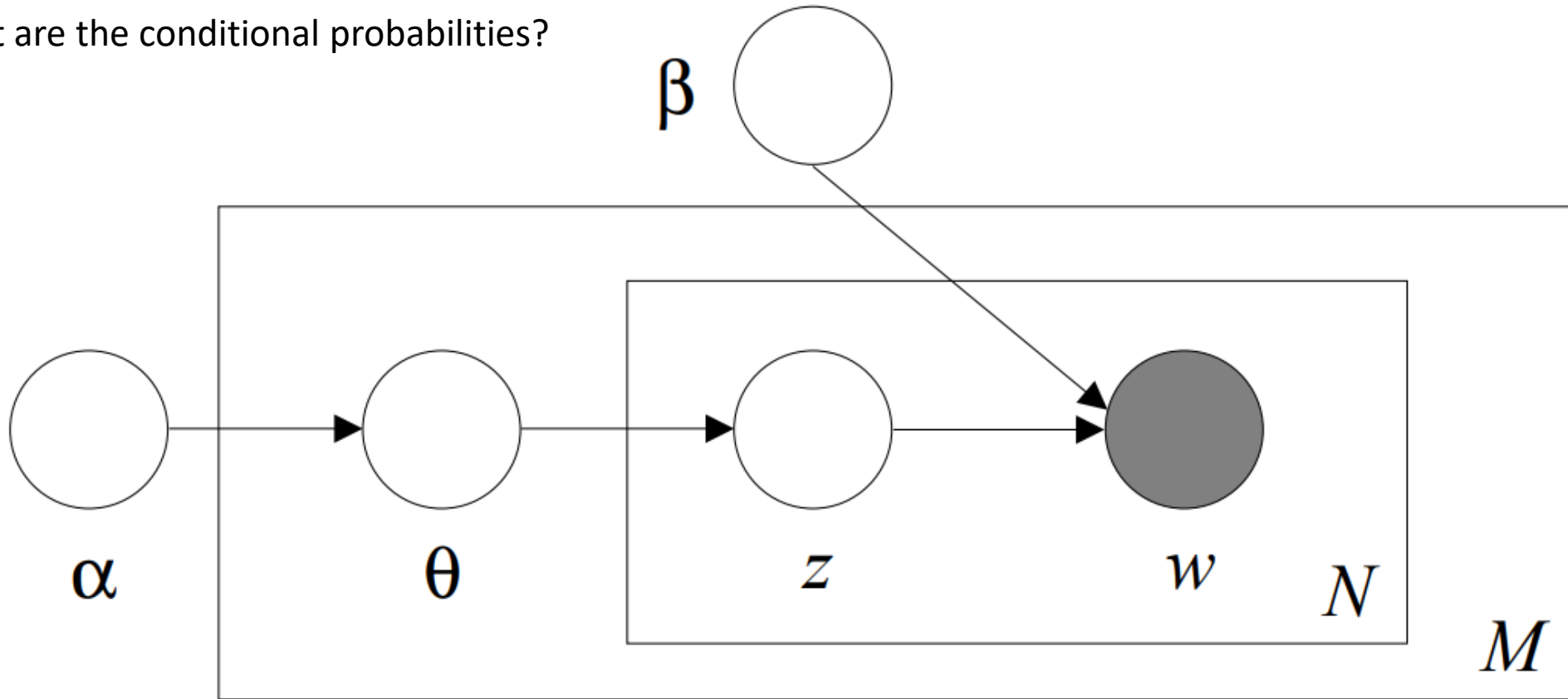
$$p(Y = 1 \mid x_1, \dots, x_n) = \frac{p(Y = 1) \prod_{i=1}^n p(x_i \mid Y = 1)}{\sum_{y=\{0,1\}} p(Y = y) \prod_{i=1}^n p(x_i \mid Y = y)}$$

Example: naïve Bayes Classifier

- Simple approach for classification
 - Computationally efficient
 - Easy to define
- Performs well in domains with many weakly relevant features
- Performs poorly when many features are strongly correlated due to the strong independence assumptions

Latent Dirichlet allocation

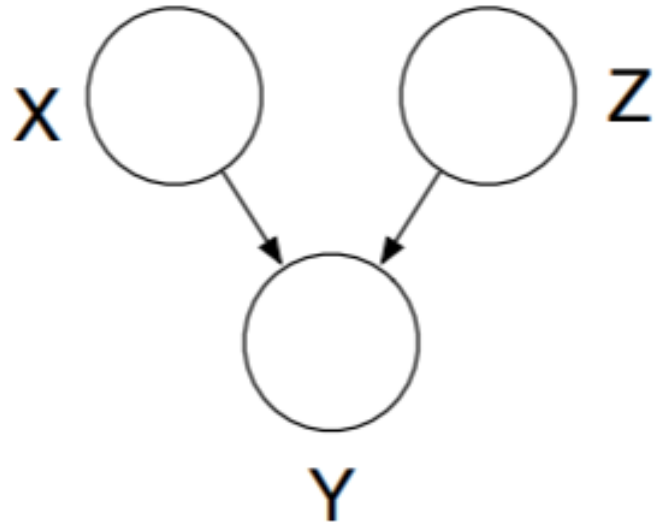
What are the conditional probabilities?



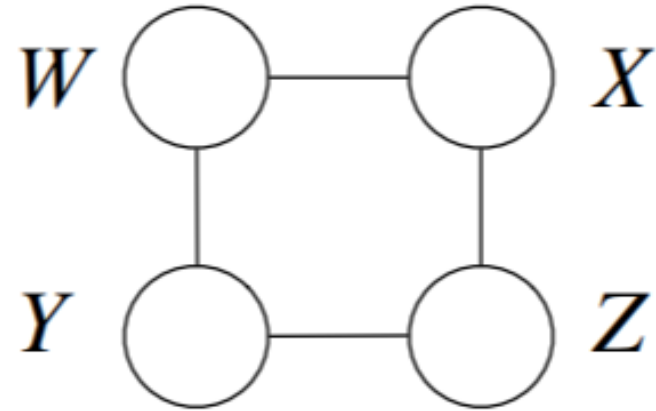
Markov random fields

- Undirected graphical models
- Consider $G = (V, E)$ and three sets of nodes A, B, C
- Separability is defined in the naïve way
 - What is an algorithm for finding conditional independencies?
- We want a joint that $X_A \perp\!\!\!\perp X_C \mid X_B$ if X_B separates X_A and X_C , meaning if every path from X_A to X_C goes through a node in X_B then we can make this assertion
- We define MRF in the reverse order (conditionals to joint)

Comparing Bayes nets and MRF



We cannot write an undirected graphical model such that this is the only conditional independence statement



The only conditional independencies implied here are below. We cannot write a directed graphical model such that these are the only conditional independence statements.

$$X \perp\!\!\!\perp Y \mid \{W, Z\}$$

$$W \perp\!\!\!\perp Z \mid \{X, Y\}$$

From local to global

- Now we derive the joint from local “functions” called potential functions

$$p(x_1, \dots, x_n) = \frac{1}{Z} \prod_{S \in \mathcal{S}} \psi(x_S)$$

S are nodes, ψ are arbitrary potential functions, $\mathcal{S}$ is a collection of subsets of nodes.

Z is a normalizing constant

Potential functions

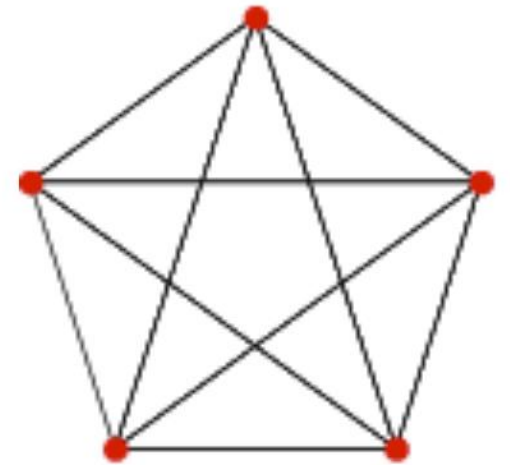
- We want the potential functions to represent conditional independencies.
- E.g., if X_1 and X_3 are separated by X_2 , this implies

$$p(x_1, x_3 | x_2) = p(x_1 | x_2) p(x_3 | x_2).$$

- But these three nodes cannot participate in a single potential function because it would be impossible for the conditional to factorize across both x_1 and x_3

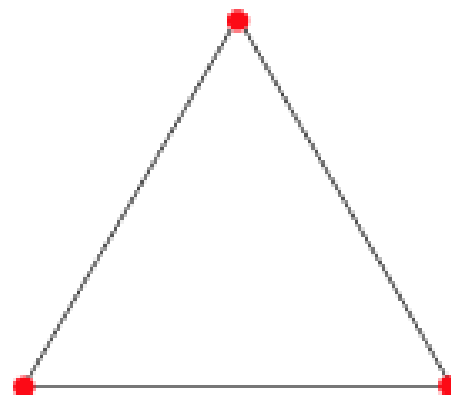
What about pairwise potentials?

- We can define a potential function for every pair of vertices.
- Assume n random variables each with a domain of d values. How many parameters can we have (entries in table)?
- What about one factor for K_n (K_5 shown to the right)?
- Different factor representations for the same graph!

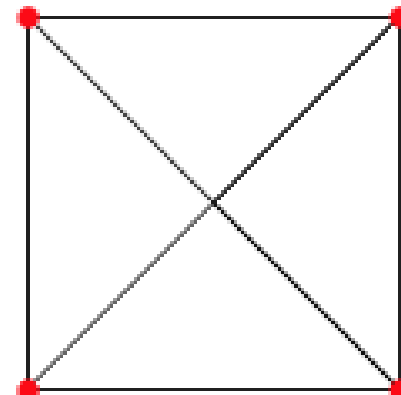




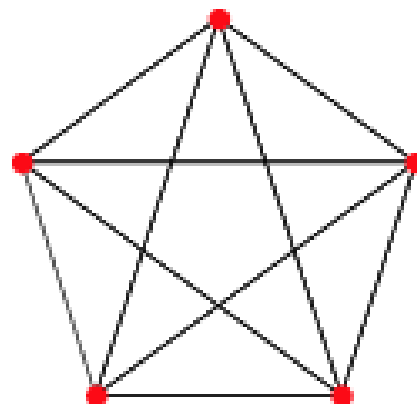
K_2



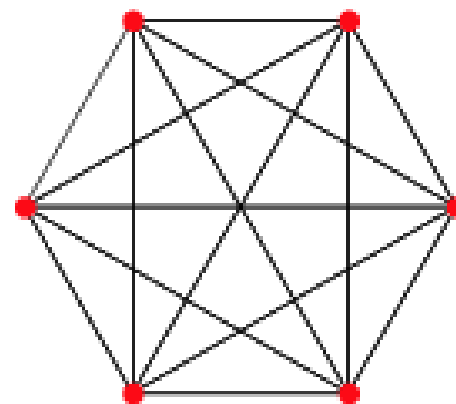
K_3



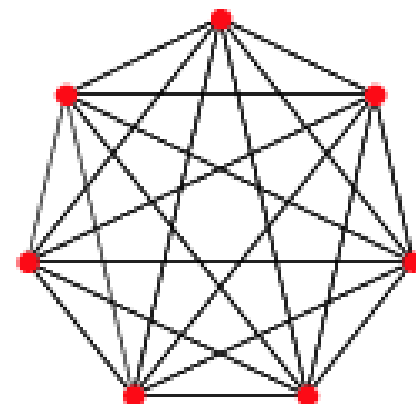
K_4



K_5



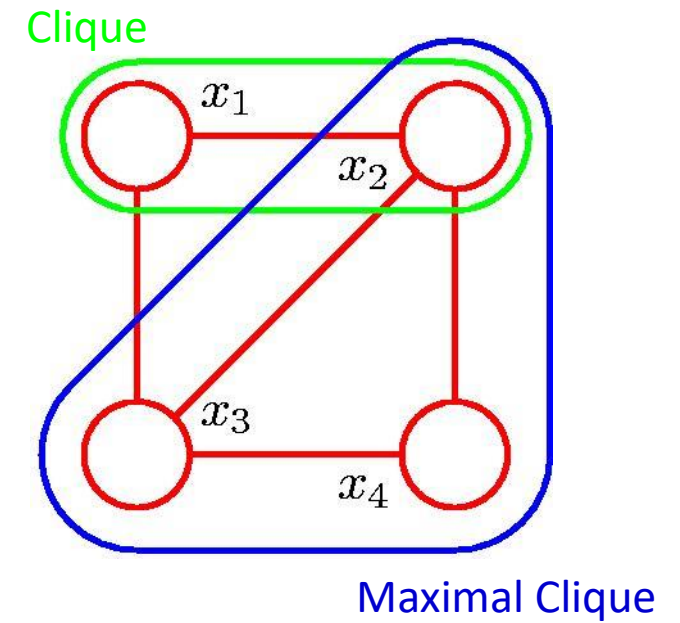
K_6



K_7

Maximal cliques

- suggestion: potentials can be defined for cliques
- All nodes are fully connected
 - conditional independencies are obeyed

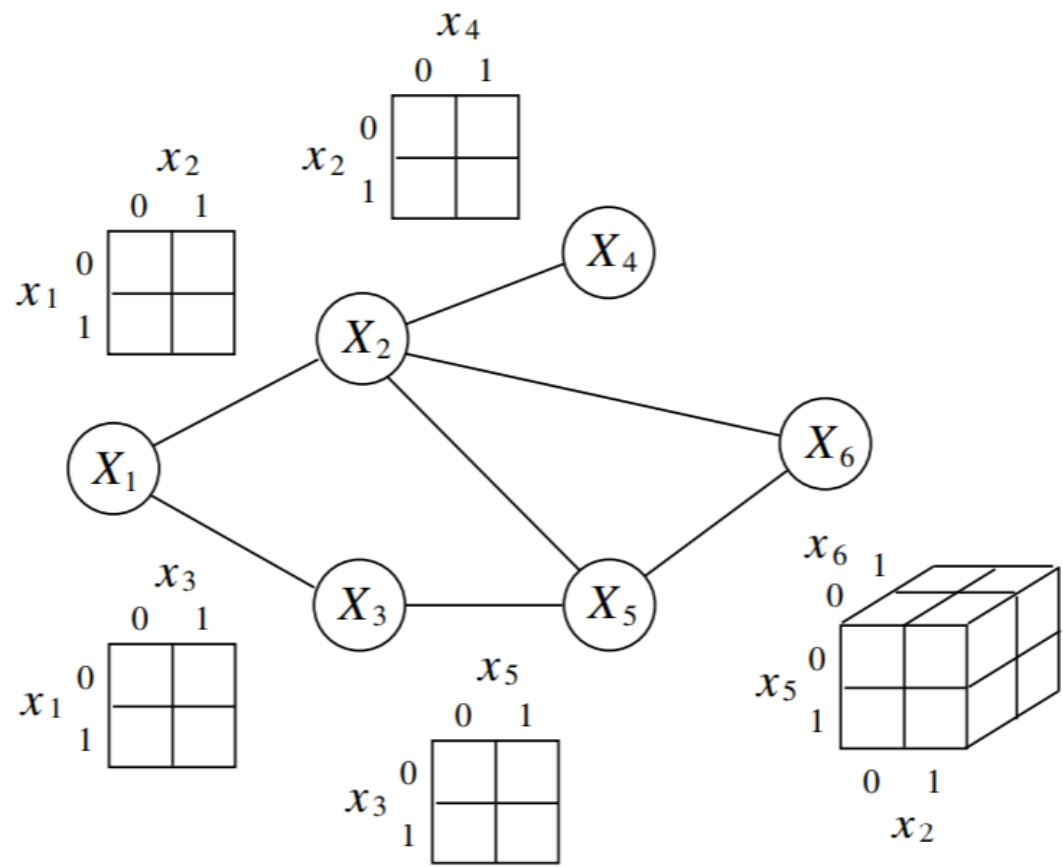


Gibbs Distribution

$$p(x) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \psi(x_C).$$

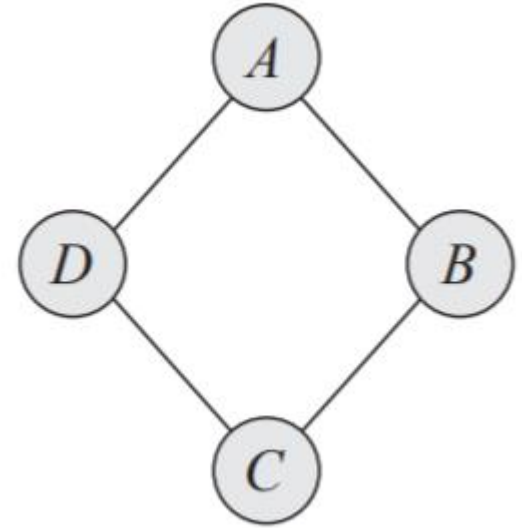
$$Z = \sum_x \prod_{C \in \mathcal{C}} \psi(x_C).$$

Maximal cliques



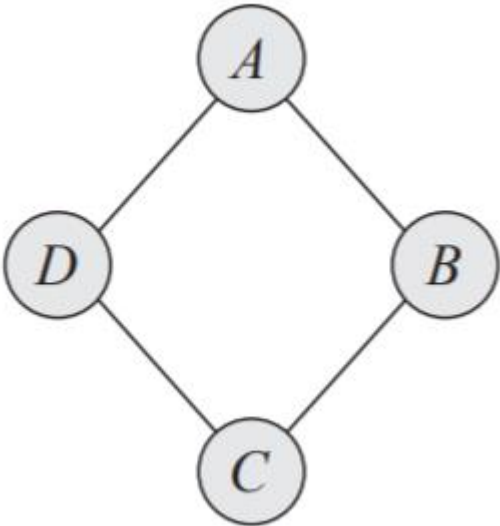
Misconception example

- 4 students, work in pairs
- Misconception about a reading by a subset
- How to parameterize?



Misconception example

- 4 students, work in pairs
- Misconception about a reading by a subset
- How to parameterize?



$\phi_1(A, B)$			$\phi_2(B, C)$			$\phi_3(C, D)$			$\phi_4(D, A)$					
a^0	b^0	30		b^0	c^0	100		c^0	d^0	1		d^0	a^0	100
a^0	b^1	5		b^0	c^1	1		c^0	d^1	100		d^0	a^1	1
a^1	b^0	1		b^1	c^0	1		c^1	d^0	100		d^1	a^0	1
a^1	b^1	10		b^1	c^1	100		c^1	d^1	1		d^1	a^1	100

$$\begin{aligned}
 &A \perp\!\!\!\perp C \mid \{D, B\} \\
 &D \perp\!\!\!\perp B \mid \{A, C\}
 \end{aligned}$$

Misconception example

Assignment				Unnormalized
a^0	b^0	c^0	d^0	300,000
a^0	b^0	c^0	d^1	300,000
a^0	b^0	c^1	d^0	300,000
a^0	b^0	c^1	d^1	30
a^0	b^1	c^0	d^0	500
a^0	b^1	c^0	d^1	500
a^0	b^1	c^1	d^0	5,000,000
a^0	b^1	c^1	d^1	500
a^1	b^0	c^0	d^0	100
a^1	b^0	c^0	d^1	1,000,000
a^1	b^0	c^1	d^0	100
a^1	b^0	c^1	d^1	100
a^1	b^1	c^0	d^0	10
a^1	b^1	c^0	d^1	100,000
a^1	b^1	c^1	d^0	100,000
a^1	b^1	c^1	d^1	100,000

$$P(a,b,c,d) = \frac{1}{Z} \phi_1(a,b) \cdot \phi_2(b,c) \cdot \phi_3(c,d) \cdot \phi_4(d,a),$$

$$Z = \sum_{a,b,c,d} \phi_1(a,b) \cdot \phi_2(b,c) \cdot \phi_3(c,d) \cdot \phi_4(d,a)$$

Misconception example

Assignment				Unnormalized	Normalized
a^0	b^0	c^0	d^0	300,000	0.04
a^0	b^0	c^0	d^1	300,000	0.04
a^0	b^0	c^1	d^0	300,000	0.04
a^0	b^0	c^1	d^1	30	$4.1 \cdot 10^{-6}$
a^0	b^1	c^0	d^0	500	$6.9 \cdot 10^{-5}$
a^0	b^1	c^0	d^1	500	$6.9 \cdot 10^{-5}$
a^0	b^1	c^1	d^0	5,000,000	0.69
a^0	b^1	c^1	d^1	500	$6.9 \cdot 10^{-5}$
a^1	b^0	c^0	d^0	100	$1.4 \cdot 10^{-5}$
a^1	b^0	c^0	d^1	1,000,000	0.14
a^1	b^0	c^1	d^0	100	$1.4 \cdot 10^{-5}$
a^1	b^0	c^1	d^1	100	$1.4 \cdot 10^{-5}$
a^1	b^1	c^0	d^0	10	$1.4 \cdot 10^{-6}$
a^1	b^1	c^0	d^1	100,000	0.014
a^1	b^1	c^1	d^0	100,000	0.014
a^1	b^1	c^1	d^1	100,000	0.014

$$P(b^1) \approx 0.732$$

$$P(b^1 \mid c^0) \approx 0.06$$

Potential functions notation

- Potential functions favor certain local configurations of variables by assigning them larger values
- They are normalized to form a proper joint distribution

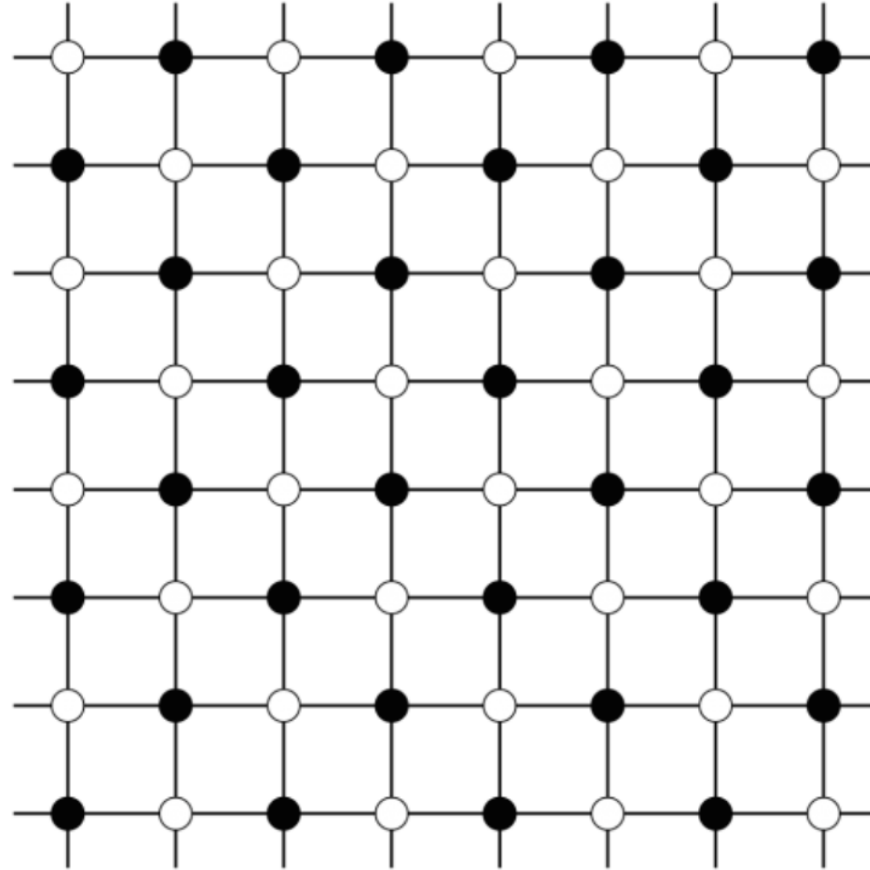
$$\psi_{X_C}(x_C) = \exp\{-H_C(x_C)\},$$

$$\begin{aligned} p(x) &= \frac{1}{Z} \prod_{C \in \mathcal{C}} \exp\{-H_C(x_C)\} \\ &= \frac{1}{Z} \exp\left\{-\sum_{C \in \mathcal{C}} H_C(x_C)\right\} \end{aligned}$$

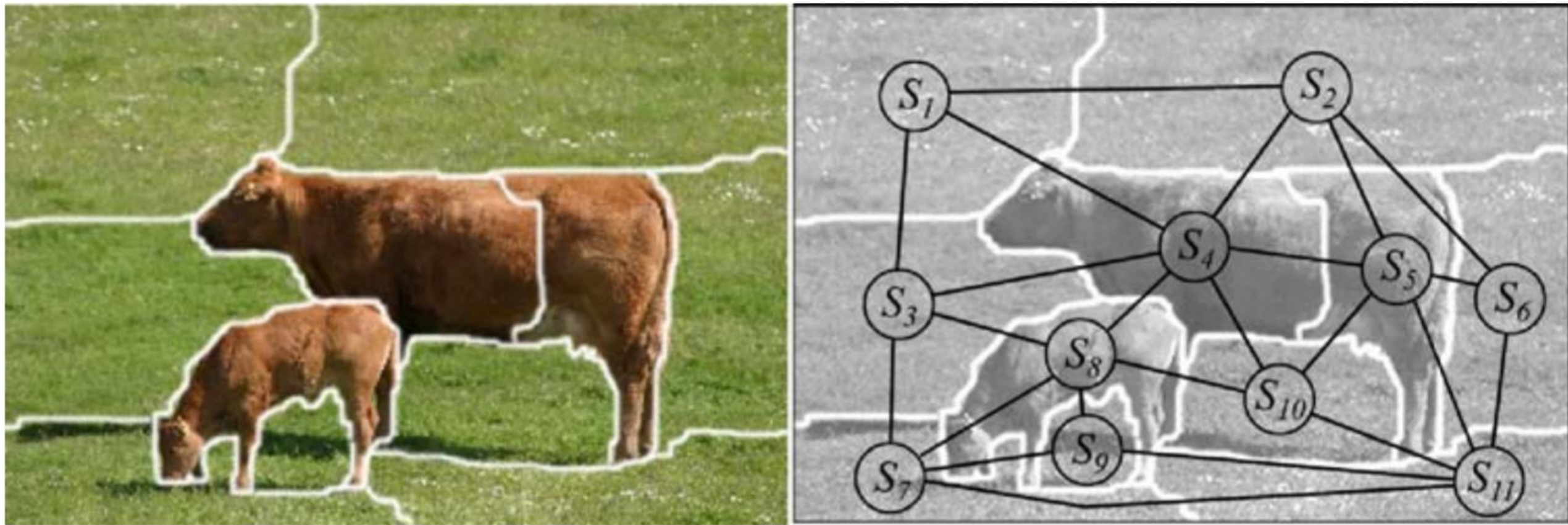
$$H(x) \triangleq \sum_{C \in \mathcal{C}} H_C(x_C)$$

$$p(x) = \frac{1}{Z} \exp\{-H(x)\}.$$

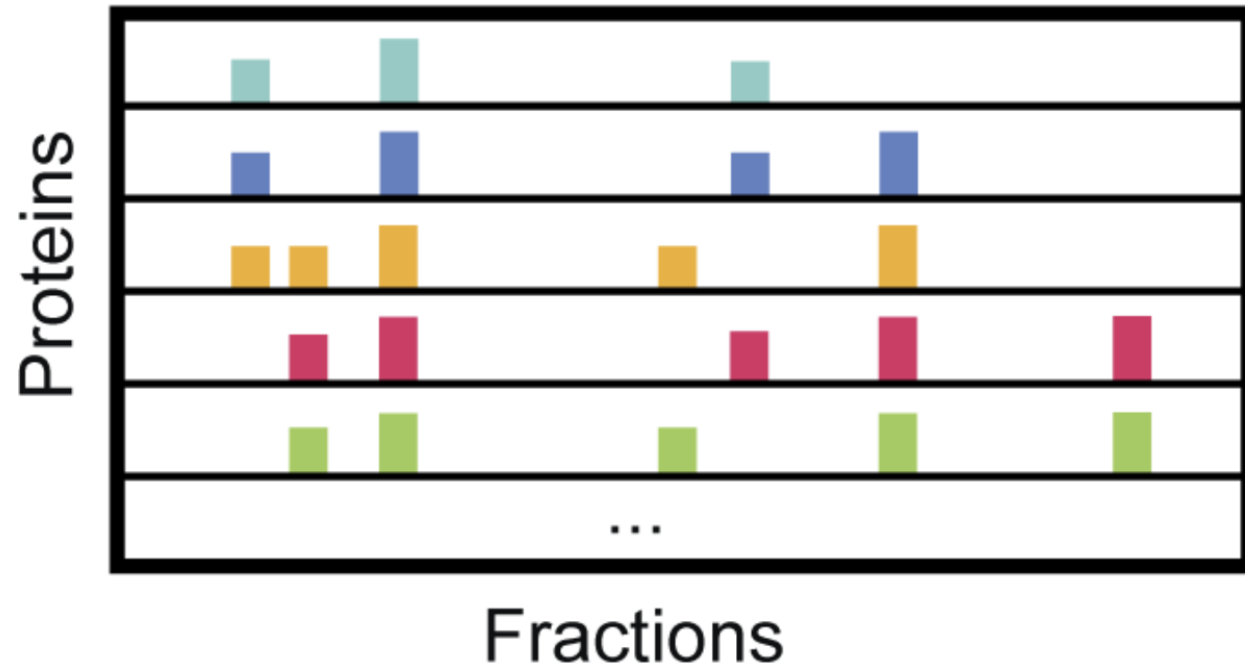
Markov Networks



Markov Networks



Protein networks



Questions?

